بسم الله الرحمن الرحيم

«وما أوتيتم من العلم الا قليلا»



Suppurative Lung Diseases

By

Mohammad Gamal Abdalrahman

Lecturer of Chest diseases

Faculty of Medicine

Assiut University

Suppurative Lung Diseases

They are group of pulmonary disorders that characterized by presence of pus within the lung parenchyma.

1- Bronchiectasis

2- Lung abscess

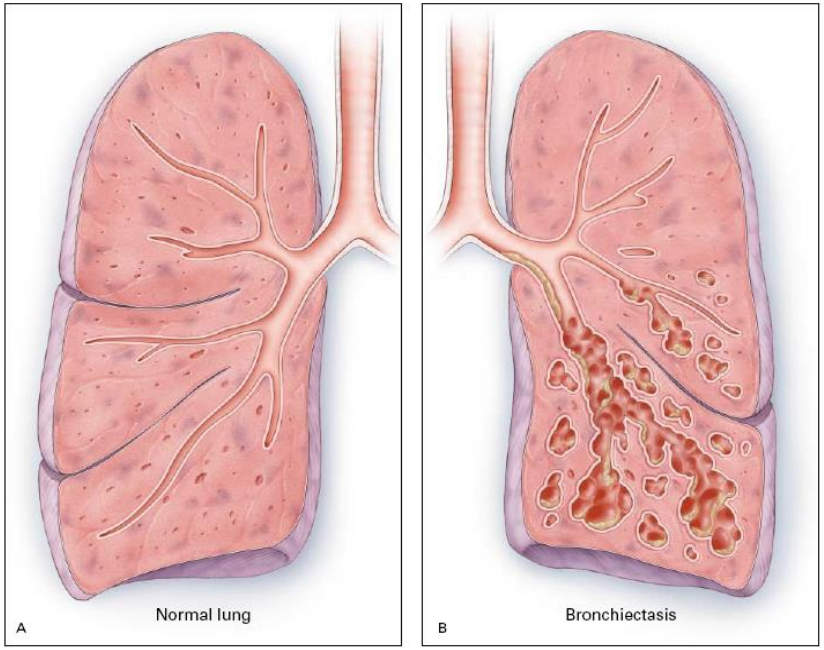
3- Empyema with broncho-pleural fistula

4- Infected cystic lung

Bronchiectasis

Bronchiectasis

Definition



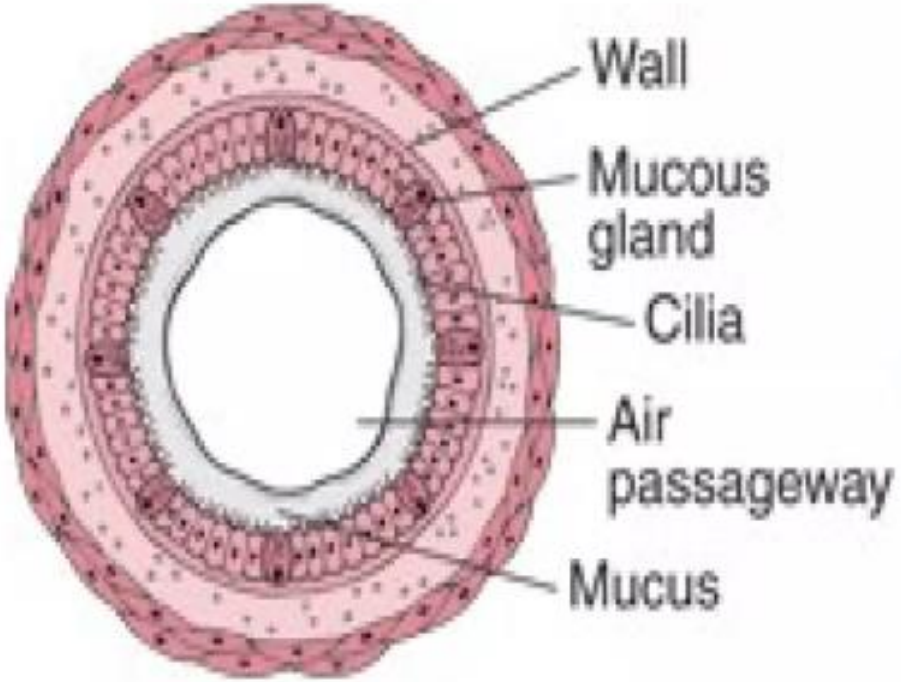
Bronchiectasis is **abnormal permanent dilatation** of one or more bronchi or bronchioles, caused by **destruction** of muscular and elastic components of the bronchial walls.

Most often secondary to an **infectious** process.

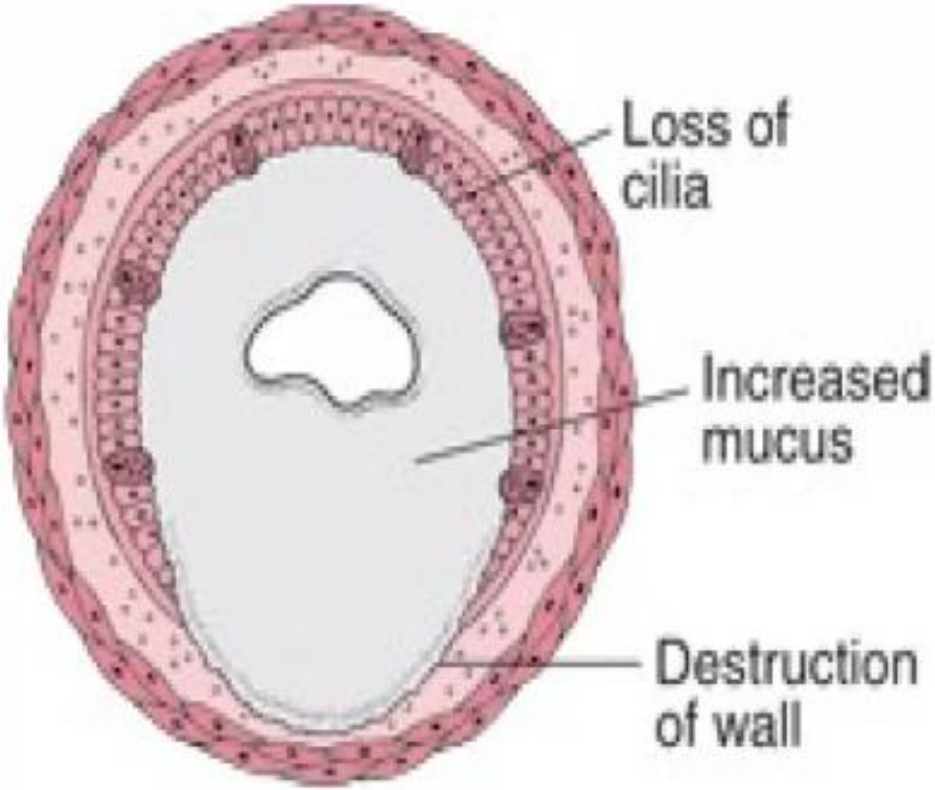
Definition

Irreversible dilatation of the cartilage containing airways

- Bronkos + Ectasia = Bronchi + Dilatation

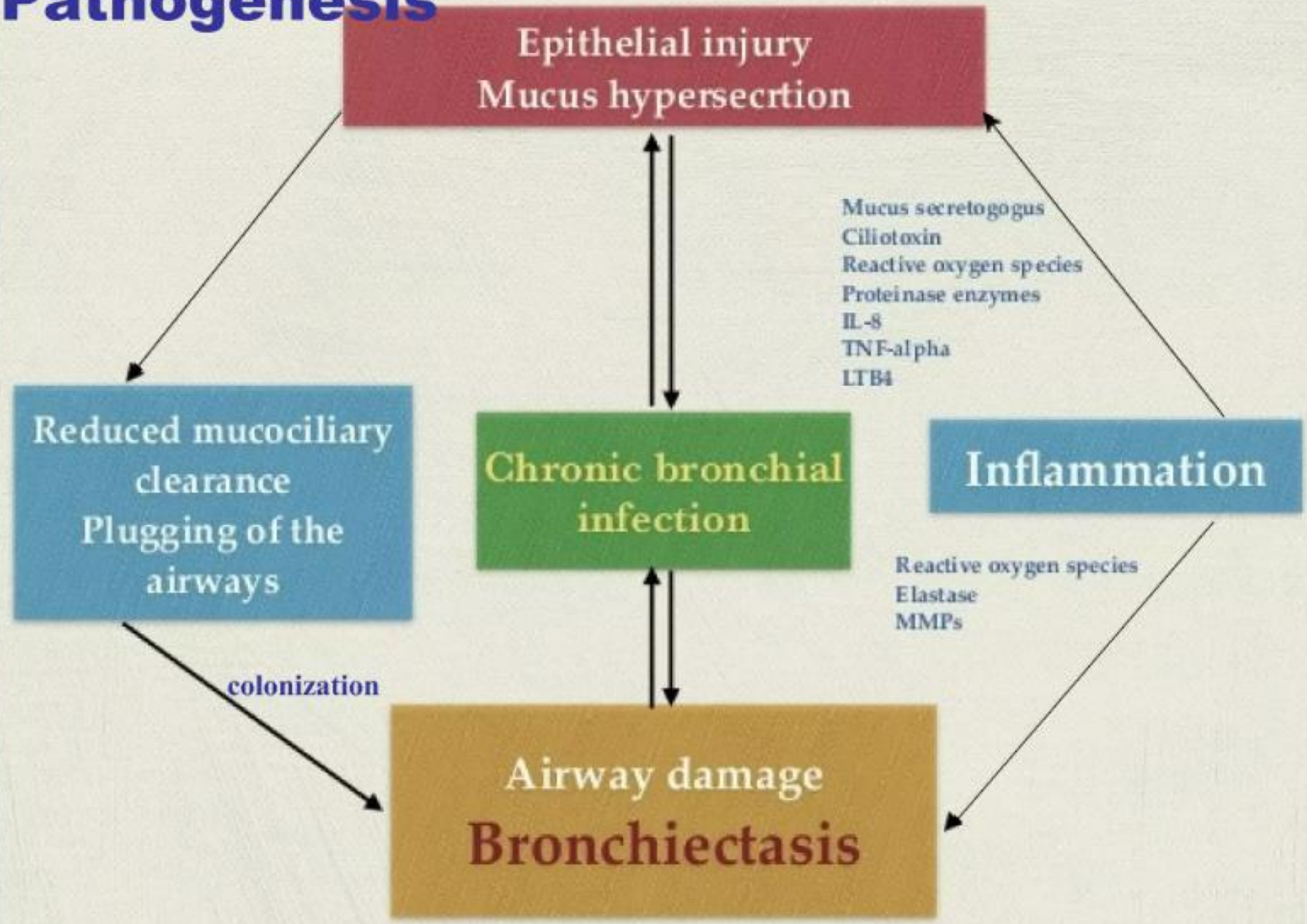


Normal Bronchus



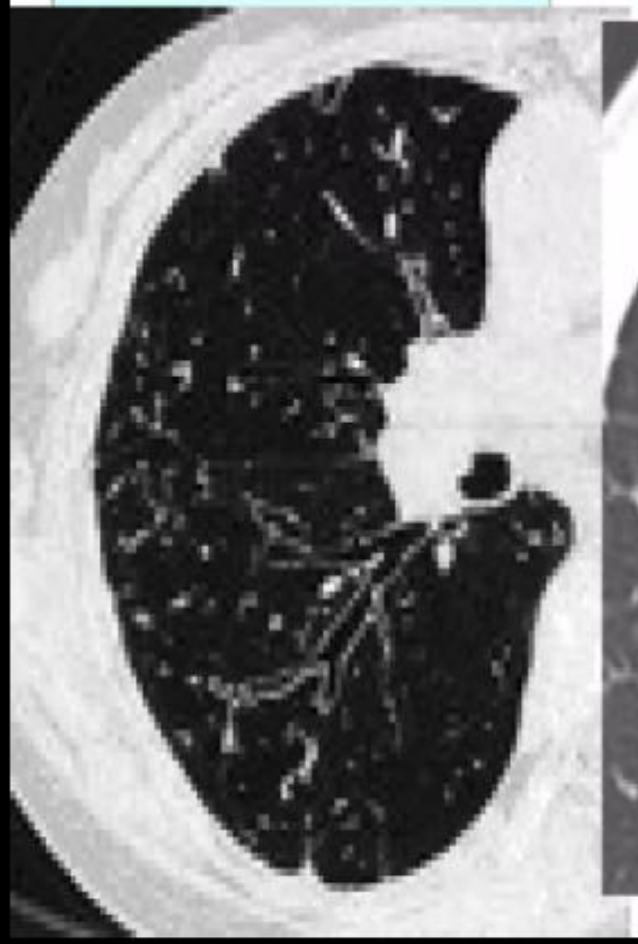
Bronchiectasis

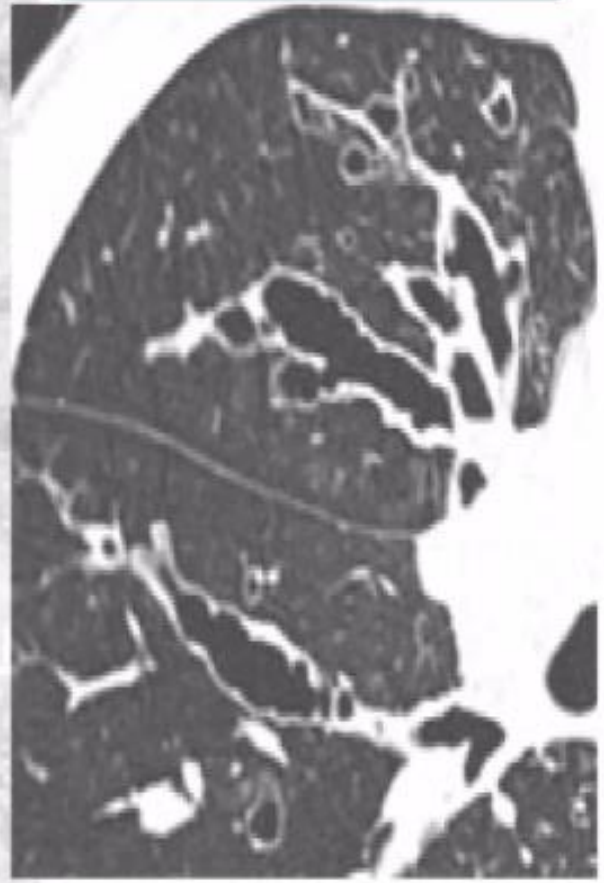
Pathogenesis

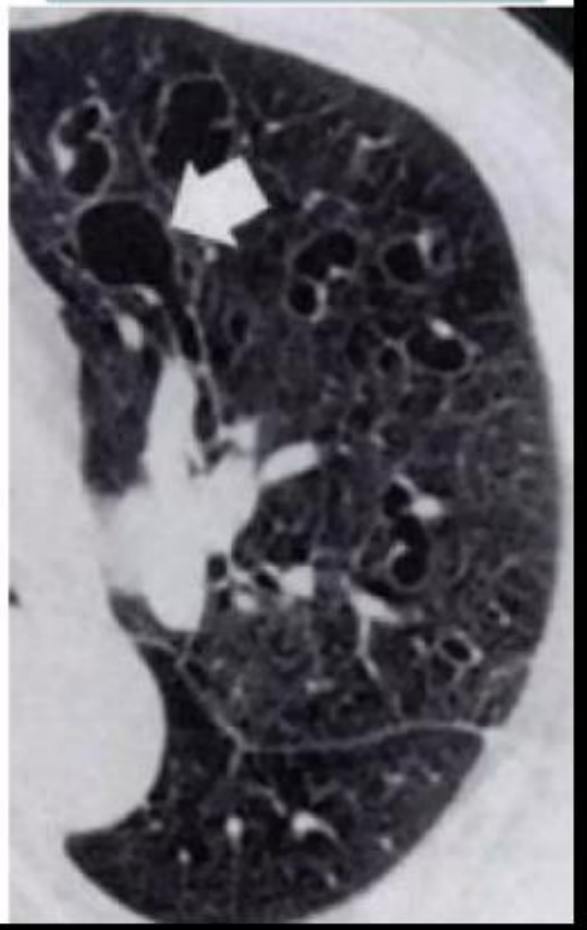


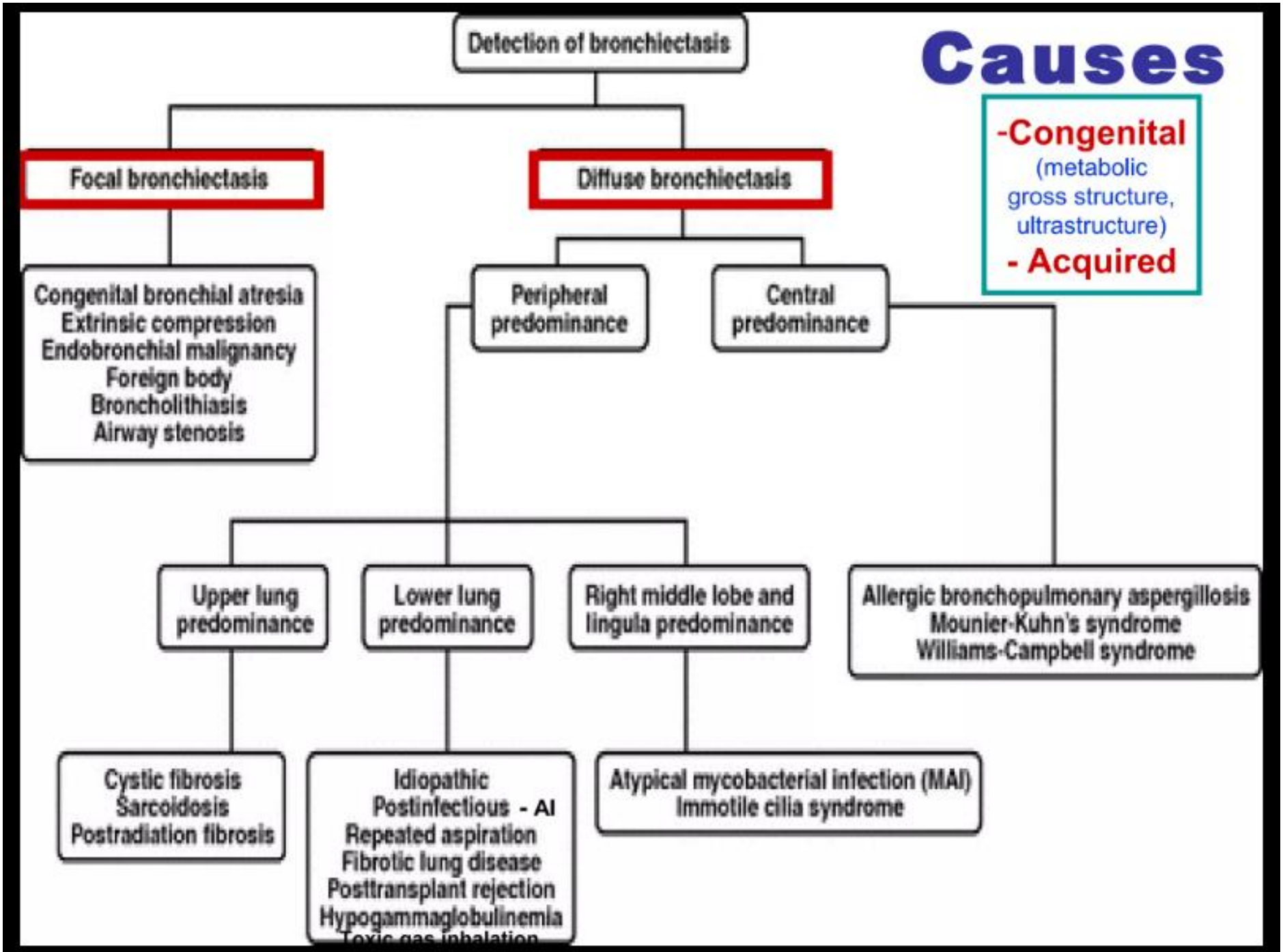
Types of bronchiectasis

- Cylindrical
- Varicose
- Cystic

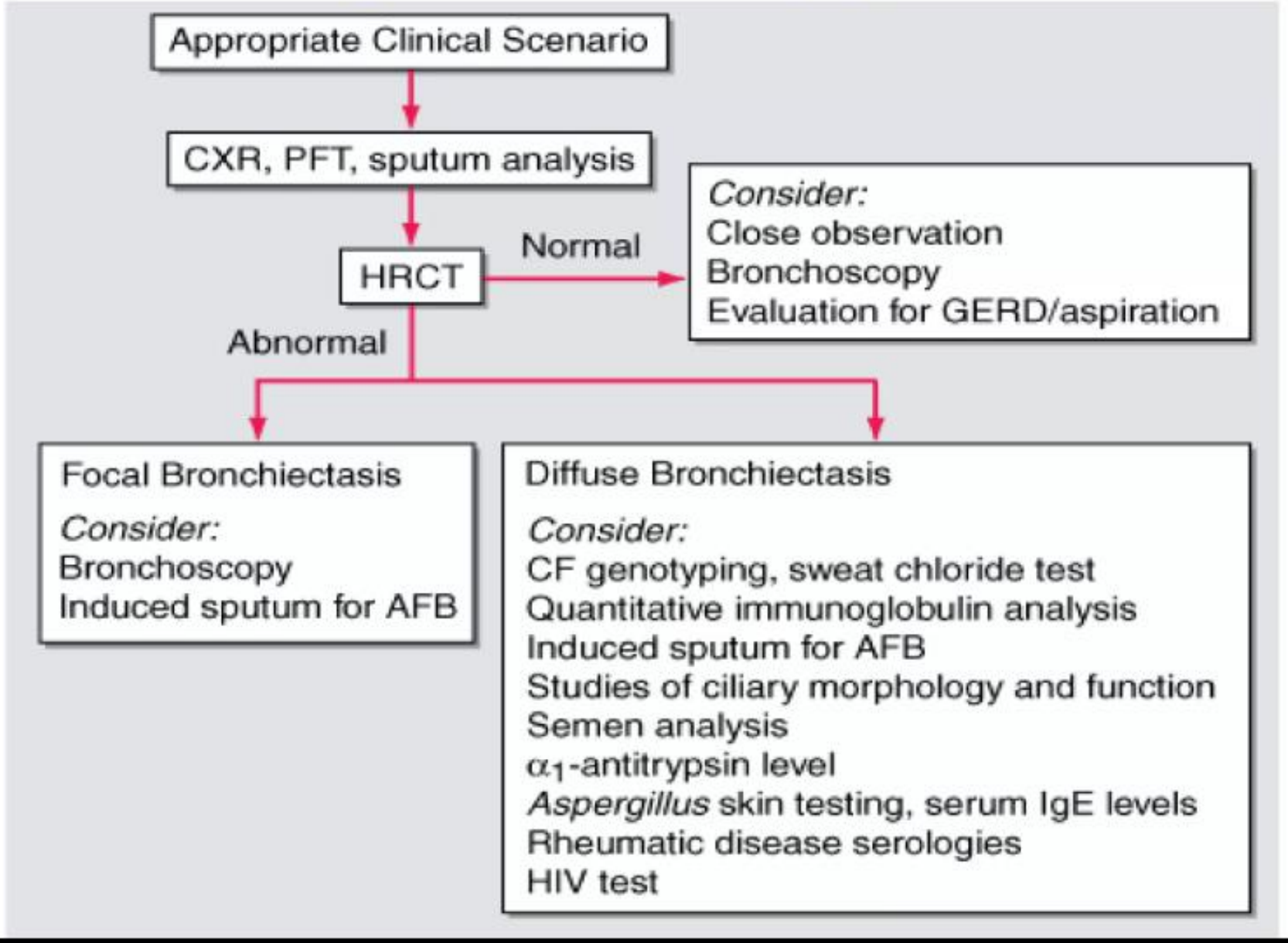







Causes
-Congenital
(metabolic
gross structure,
ultrastructure)
- Acquired
Detection of bronchiectasis
Focal bronchiectasis
Diffuse bronchiectasis
Congenital bronchial atresia
Extrinsic compression
Endobronchial malignancy
Foreign body
Broncholithiasis
Airway stenosis
Peripheral predominance
Central predominance
Upper lung predominance
Lower lung predominance
Right middle lobe and lingula predominance
Allergic bronchopulmonary aspergillosis
Mounier-Kuhn's syndrome
Williams-Campbell syndrome
Cystic fibrosis
Sarcoidosis
Postradiation fibrosis
Idiopathic
Postinfectious - AI
Repeated aspiration
Fibrotic lung disease
Posttransplant rejection
Hypogammaglobulinemia
Toxic gas inhalation
Atypical mycobacterial infection (MAI)
Immotile cilia syndrome

Diagnostic Approach



Clinical Features:

- Cough
- Daily sputum production: green/yellow sputum (patients with bronchiectasis may produce 240ml (8 oz) of sputum daily).
- Dyspnea
- Wheezing
- Hemoptysis
- Bluish skin color
- Recurrent pleurisy
- Dry Bronchiectasis:
- Breath odor
- Clubbing of fingers
- Fatigue
- Paleness
- Weight loss
- Acute exacerbation: ↑ purulence, amount of Sputum & Dyspnea
- Late: RF, Corpulmonale



Complications:

- Recurrent hemoptysis, pneumonia and pleurisy
- Lung abscess and metastatic brain abscess
- Empyema
- Amyloidosis
- Cor pulmonale and respiratory failure

- **CXR:**

Suspicious but not diagnostic radiographic findings include:

focal pneumonitis, scattered irregular opacities that may represent mucopurulent plugs, linear or plate-like atelectasis , dilated and thickened airways that appear as ring-like shadows (of airways that are seen on end) or tram lines (airways that are perpendicular to the x-ray beam)

- **HRCT:**

Major features on HRCT include AW dilatation & bronchial wall thickening

- **Bronchoscopy:**

For diagnosis of tumor, foreign body, localize site of hemoptysis.

- **PFTs**

- **Blood** (including serology) **& Sputum Examination**

Sputum Organisms

:Typical offending organisms

- Klebsiella species,
- Staphylococcus aureus,
- Mycobacterium tuberculosis,
- Mycoplasma pneumoniae,
- Non-tuberculous mycobacteria,
- Measles virus,
- Pertussis virus,
- Influenza virus,
- Herpes simplex virus, and
- Certain types of adenovirus.
- Aspergillus fumigatus

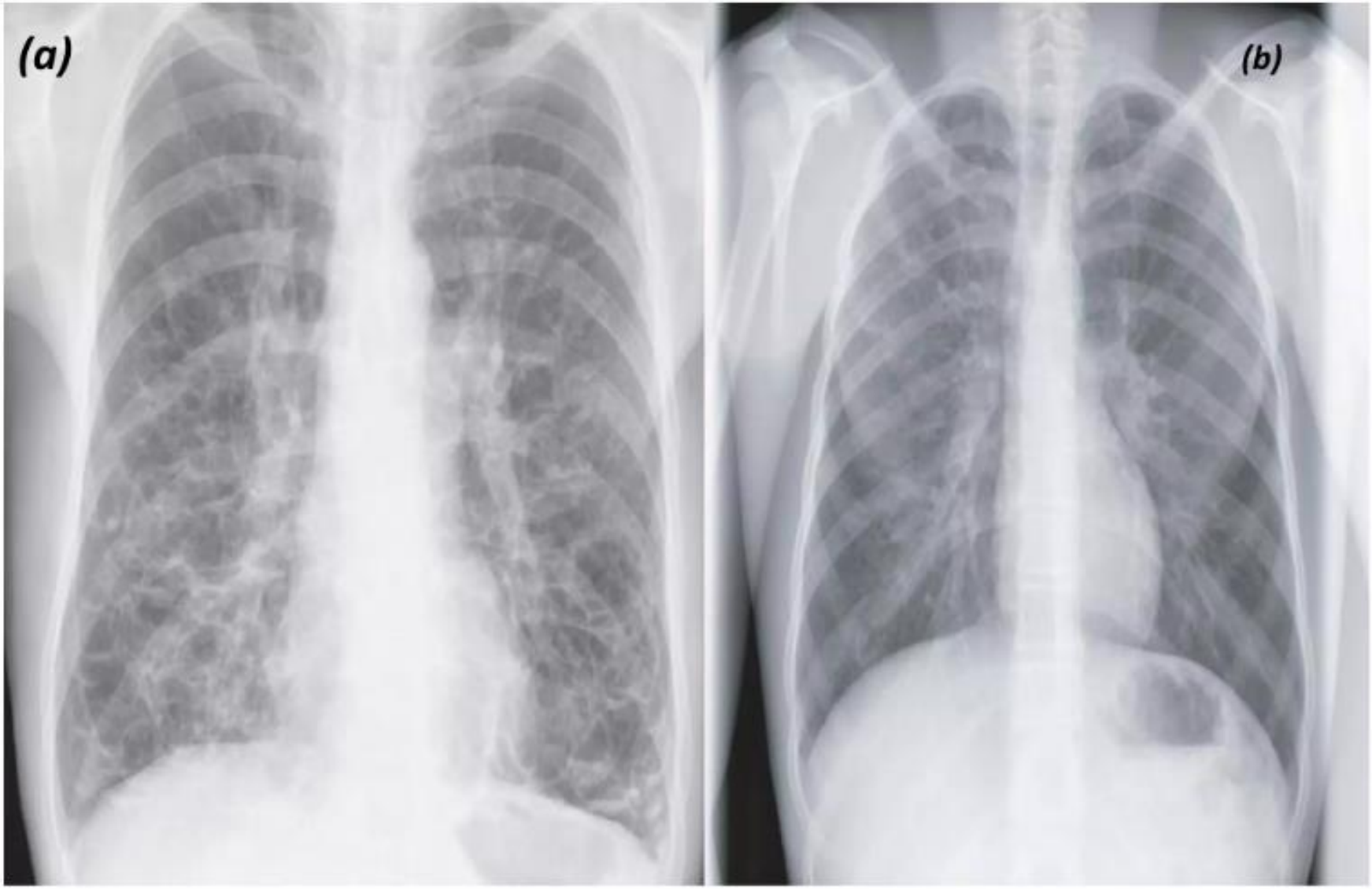
Once a patient develops bronchiectasis:

- Many of these same organisms colonize the damaged bronchi and may result in ongoing damage and episodic infectious exacerbations.

- The organisms found most typically include ***Haemophilus*** **species and** ***Pseudomonas*** **species (Staph Aureus in CF)**

Plain radiographic signs

Normal	•In most cases.
Specific findings of bronchial dilatation	•Tram trak shadows •Signet ring shadows
Non specific findings	• Peribronchial cuffing (thickened hazy bronchial wall). • Finger in glove opacities (mucus filled bronchi). • Multiple air fluid levels (fluid filled bronchi).



Chest radiography showing a) cystic bronchiectasis with multiple cystic airspaces and b) cylindrical bronchiectasis and tram track opacities in a cystic fibrosis patient.



Cystic bronchiectasis with air-fluid levels.

CT signs

Specific findings of bronchial dilatation

- Signet ring sign (broncho-arterial ratio>1).
- Lack of bronchial tapering.
- Abnormal bronchial contour.
- Visibility of peripheral air ways.

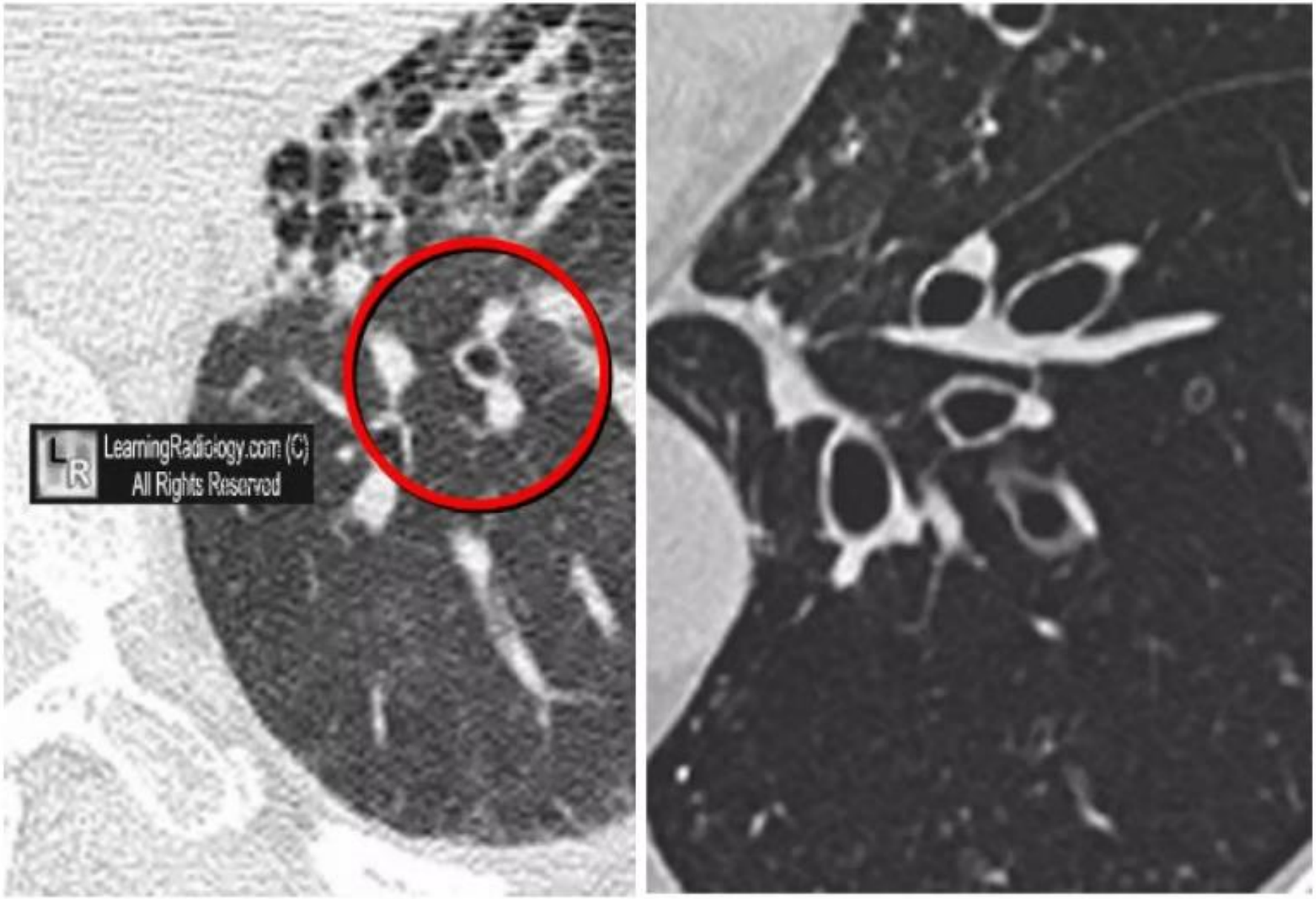
Non specific findings

- Peribronchial cuffing (thickened hazy bronchial wall).
- Finger in glove opacities (mucus filled bronchi).
- Multiple air fluid levels (fluid filled bronchi).

Ancillary findings of small air way disease

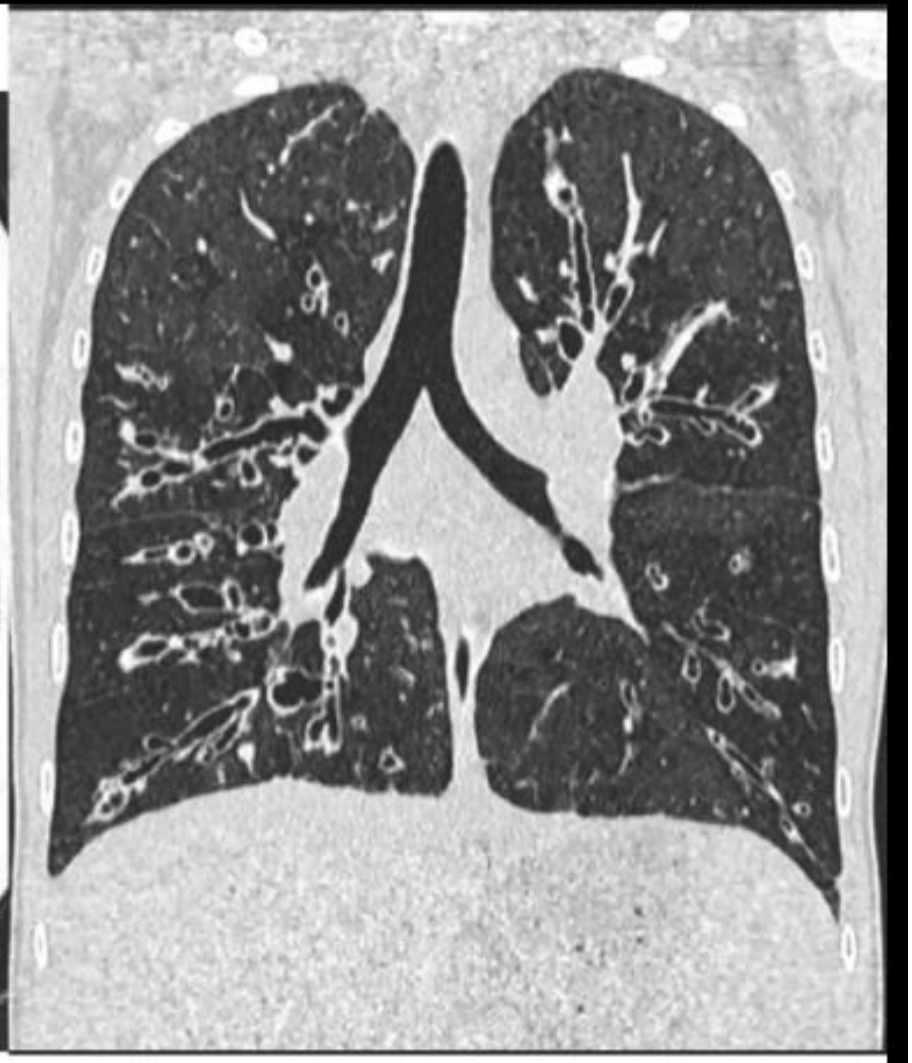
- Mosaic perfusion.
- Air trapping.
- Tree in bud opacities.

*Cause can be seen as well



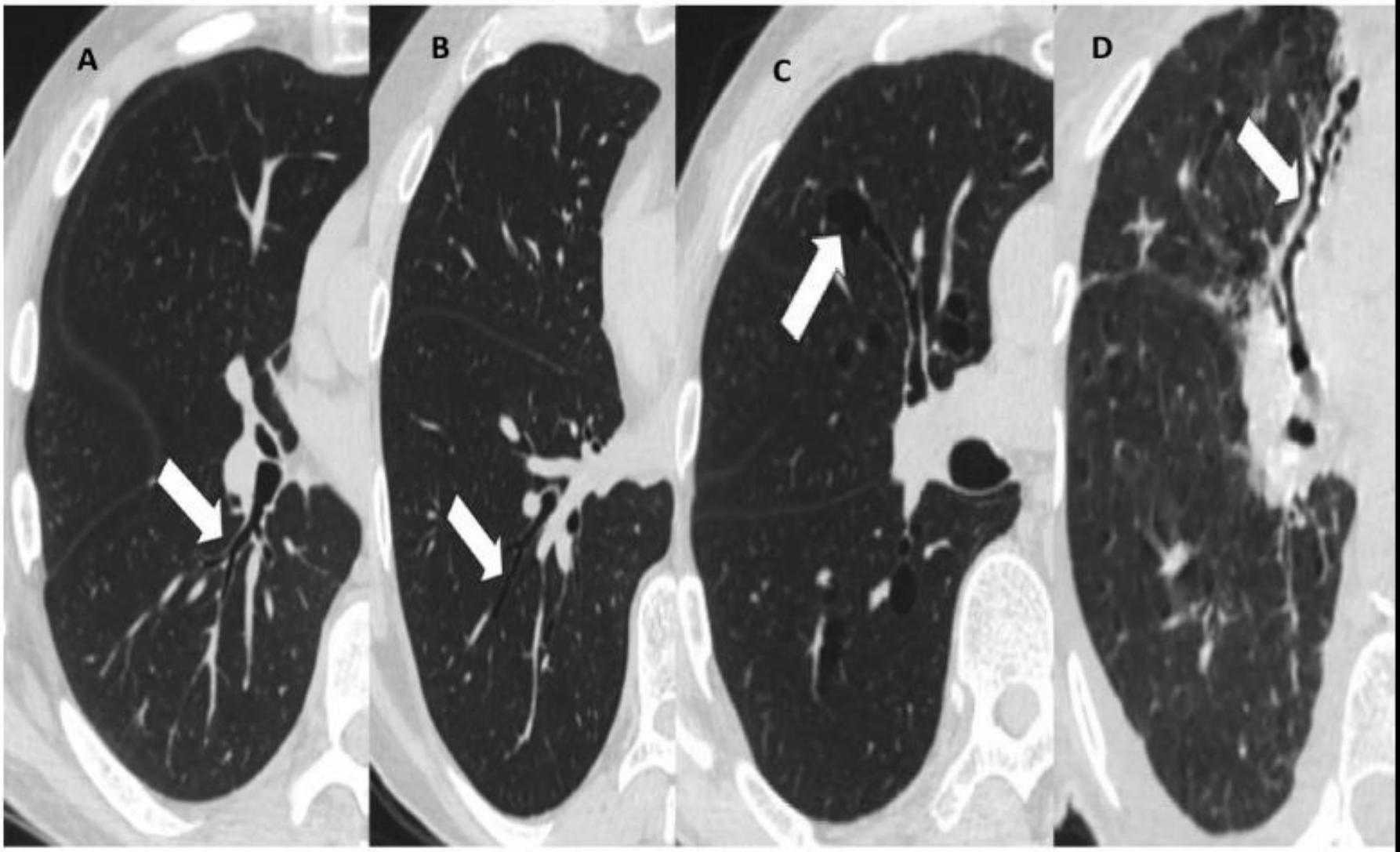
SIGNET RING SIGN. Chest CT shows small bronchiectasis.





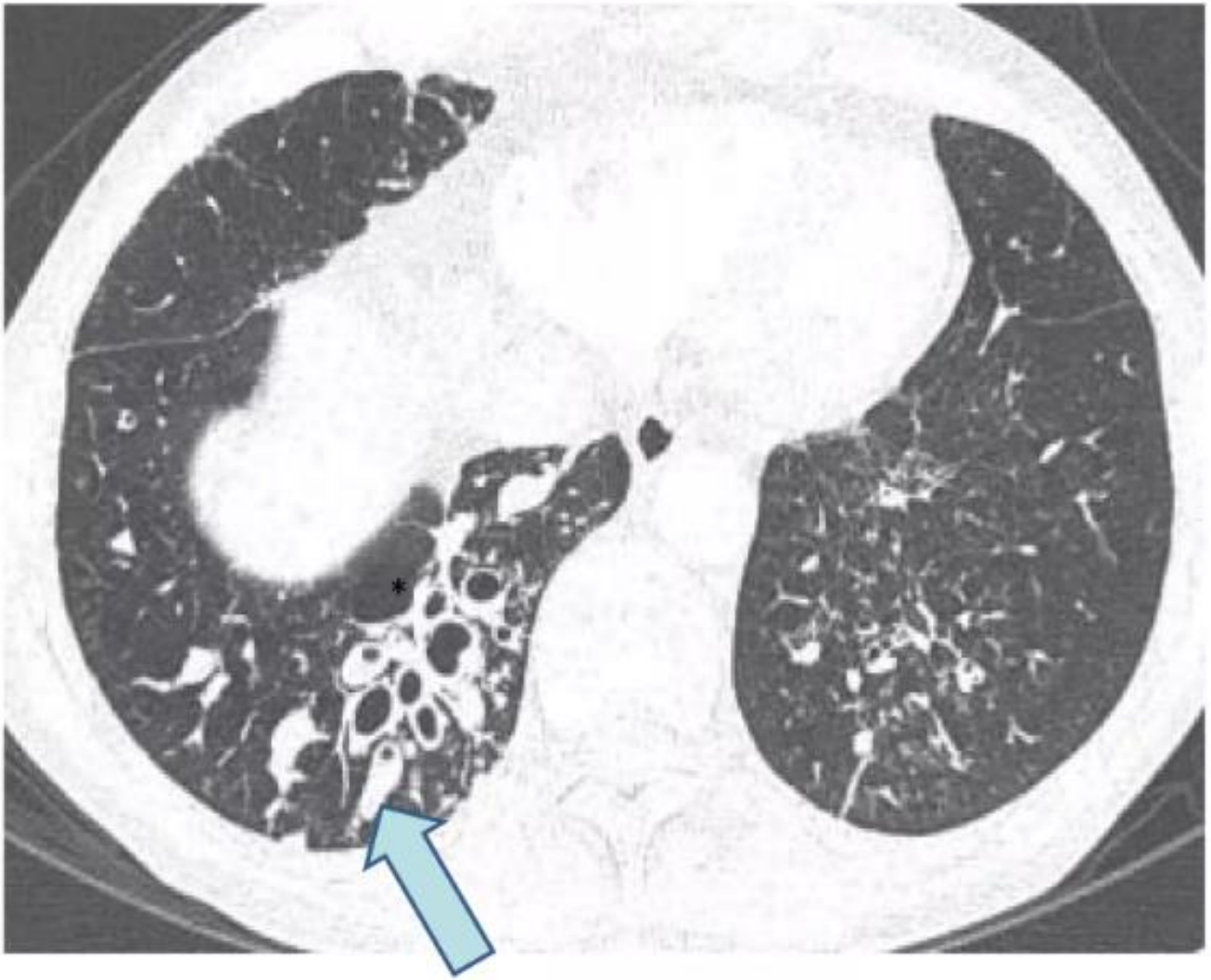
High-resolution computed tomography image showing non tapering bronchi, in keeping with bronchiectasis.

Abnormal bronchial contour.



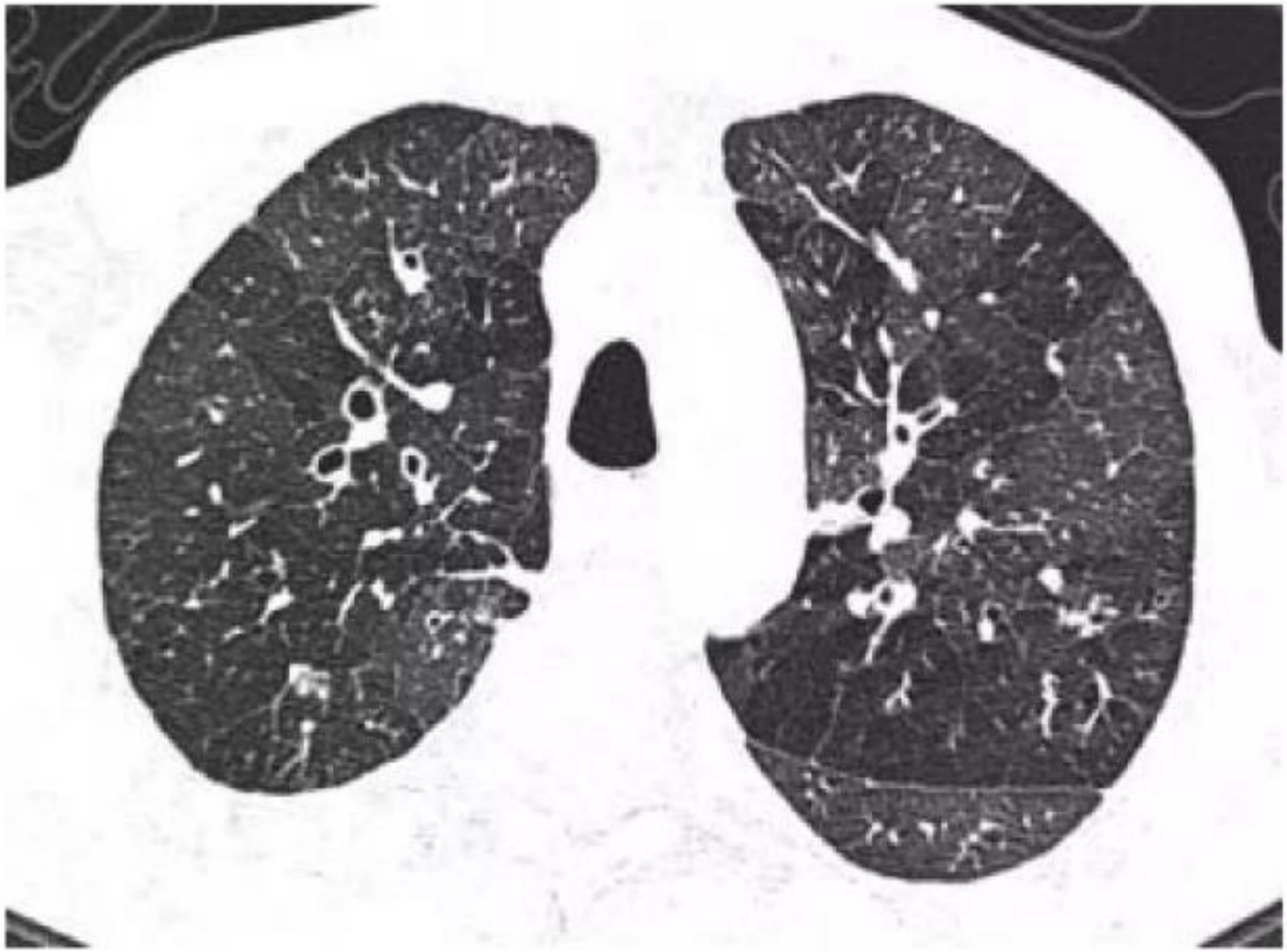
Categories of bronchiectasis. Normal bronchus (arrow) (A), cylindrical bronchiectasis with lack of bronchial tapering (arrow) (B), varicose bronchiectasis with string-of-pearls appearance (arrow) (C), and cystic bronchiectasis (arrow) (D).

Bronchial wall thickening



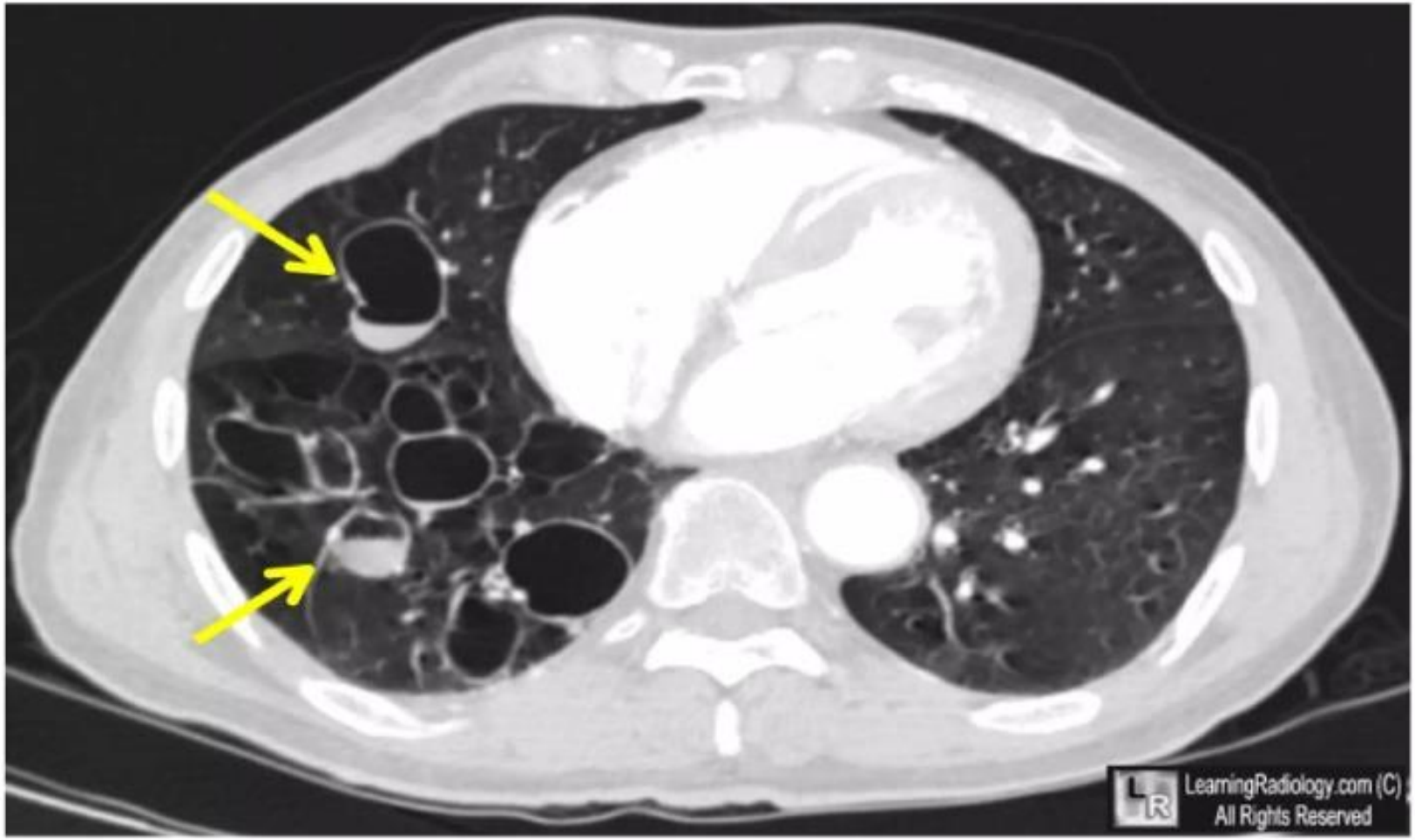
High-resolution computed tomography image demonstrating bronchiectasis with bronchial wall thickening (asterisk) and mucous plugging (arrow) in the right lower lobe.

Air-trapping Sign.



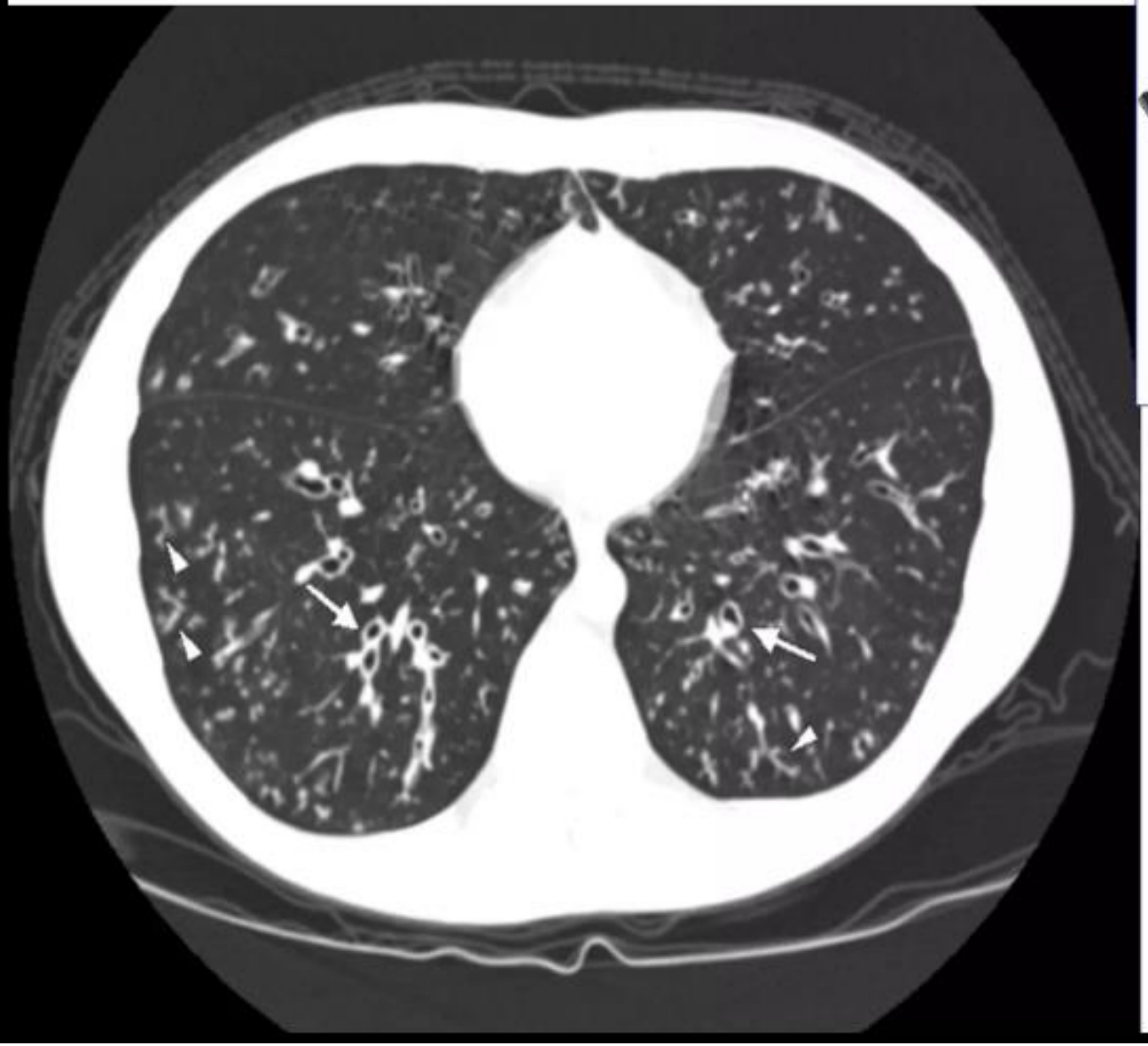
Inspiratory high-resolution computed tomography image showing bronchiectasis and widespread areas of low attenuation, representing air-trapping.

Cystic changes with air-fluid levels.



Cystic Bronchiectasis. CT: Markedly dilated bronchi are seen, some with air-fluid levels (yellow arrows), mostly in the right lung.

Tree in budd opacities





Differential Diagnosis

I) Other causes of chronic airway diseases:

- Bronchial asthma
- COPD
- Chronic bronchitis

II) Other causes of suppurative lung syndromes:

- Empyema with bronchopleural fistula
- Chronic lung abscess

III) Other causes of hemoptysis

Treatment Outlines

1- Treatment of pulmonary infections (Antibiotics)

2- Clearance of airway secretions

- Bronchodilators
- Mucolytics
- Mobilization of secretions:
  Postural drainage and Chest physiotherapy

3- Surgery

- Resection
- Transplantation

Treatment

- **Goals:**

1. Controlling infections and bronchial secretions
2. Relieving airway obstructions
3. Removal of affected portions of lung by surgical removal or artery embolization
4. Preventing complications.

- **Treatment of bronchiectasis include:**

1. The prolonged usage of **Antibiotics** to prevent detrimental infections
2. Eliminating accumulated fluid with **Bronchial Hygiene therapy + Humidification & Mucolytics** (dornase, HS, ACC) **+/- Anti-oxidants**
3. Inhaled **Bronchodilators** (in obst.) **& Steroids** (in exac., ABPA)
4. **Surgery** is used to treat localized bronchiectasis, removing obstructions, recurrent hemoptysis & exacerbations → **Transplantation**
5. **Vaccination** (S. pn, H. Inf., M., P.), **IgG** (25mg/kg /wk), **O2, Smoke Cess**
6. **Pulmonary Rehabilitation** (add ADEK, Panc enz to CF)

Antibiotic Use

During Exacerbation:

- **Mild to Moderate Exacerbation (7-10 ds):** Amoxacillin, Tetracycline, Trimethoprim-Sulfamethoxazole, New Macrolides, Cephalosporin, Quinolones
- **Moderate to Severe Exacerbation (10-14 ds):** Antipseudomonal IV (dual therapy) or MAC treatment if proved (clarithromycin, ethambutol, rifampicin, sterptomycin 18-14 Ms)

Regular Treatment for Colonization:

- **Intermittent courses** are used for 7 days and Ab- free periods of 7 days each month can be used (oral previous drugs) or
- **Long term antibiotics** for 3 to 6 Ms (macrolides)
- **Inhaled Antibiotics:** Tobramycin, Colistin, Gentamycin, Astreonam

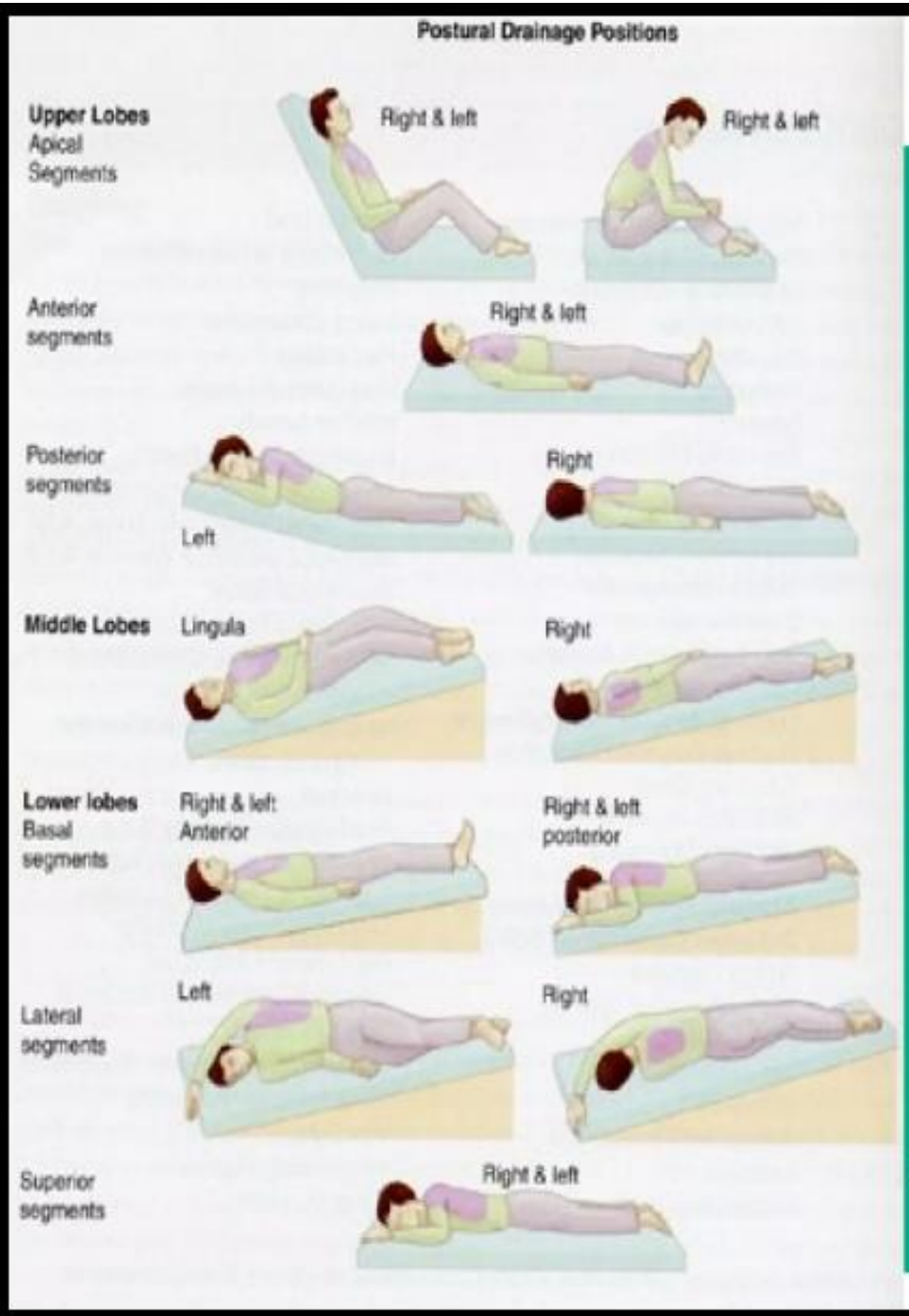
Generic Name	Brand Name	Disease Indications	Formulation	Recommended Dose/Frequency	Delivery Device	Adverse Effects
Approved use						
Aztreonam	Cayston	CF	Inhalation solution	75 mg nebulized three times a day	Altera eFlow nebulizer	Wheezing
Colistin	Colomycin	CF; ventilator-associated pneumonia	Powder dissolved in saline	1–2 million units (75–150 mg) nebulized twice a day	Jet or ultrasonic nebulizer	Cough Bronchospasm Throat irritation
	Colobreathe	CF	Dry powder	125 mg (1,662,500 IU) inhaled twice a day	Turbospin inhaler	Cough Throat irritation Abnormal taste
Tobramycin	TOBI®	CF	Inhalation solution (5-ml ampule containing 300 mg tobramycin and sodium chloride, pH 6.0)	300 mg nebulized twice a day	PARI LC PLUS jet nebulizer	Tinnitus Dysphonia Bronchospasm Hearing loss
	Bethkis	CF	Inhalation solution (4-ml ampule containing 300 mg tobramycin and sodium chloride, pH 5.0)	300 mg nebulized twice a day	PARI LC PLUS jet nebulizer	Tinnitus Dysphonia Bronchospasm Hearing loss
	TOBI Podhaler	CF	Dry powder	112 mg inhaled twice a day	T-326 inhaler	Cough
Off-label use						
Amikacin	N/A	Nontuberculous mycobacteria	Injectable solution diluted with saline	250 mg nebulized daily (up to 500 mg twice a day if tolerated)	Jet nebulizer	Hearing Loss Vertigo Dysphonia Nephrotoxicity
Amphotericin B	Fungizone	Post-transplant fungal prophylaxis or treatment	Injectable solution diluted with sterile water	Prophylaxis: 25 mg daily for 4 d then weekly for 7 wk Double dose if mechanically ventilated	Jet nebulizer	Shortness of breath Cough Abnormal taste Bronchospasm
	Abelcet	Post-transplant fungal prophylaxis or treatment	Liposomal solution	Prophylaxis: 50 mg daily for 4 d then weekly for 7 wk Double dose if mechanically ventilated	Jet nebulizer	Cough Abnormal taste Bronchospasm
Ceftazidime	Fortaz	Ventilator-associated pneumonia; CF (*Burkholderia*	Injectable solution diluted with saline	Ventilator-associated pneumonia: 15 mg/kg nebulized every 3 h	Jet nebulizer	
Tobramycin	N/A	CF	Injectable solution diluted with saline	80 mg nebulized twice a day	Jet nebulizer	Tinnitus Dysphonia Bronchospasm Hearing loss

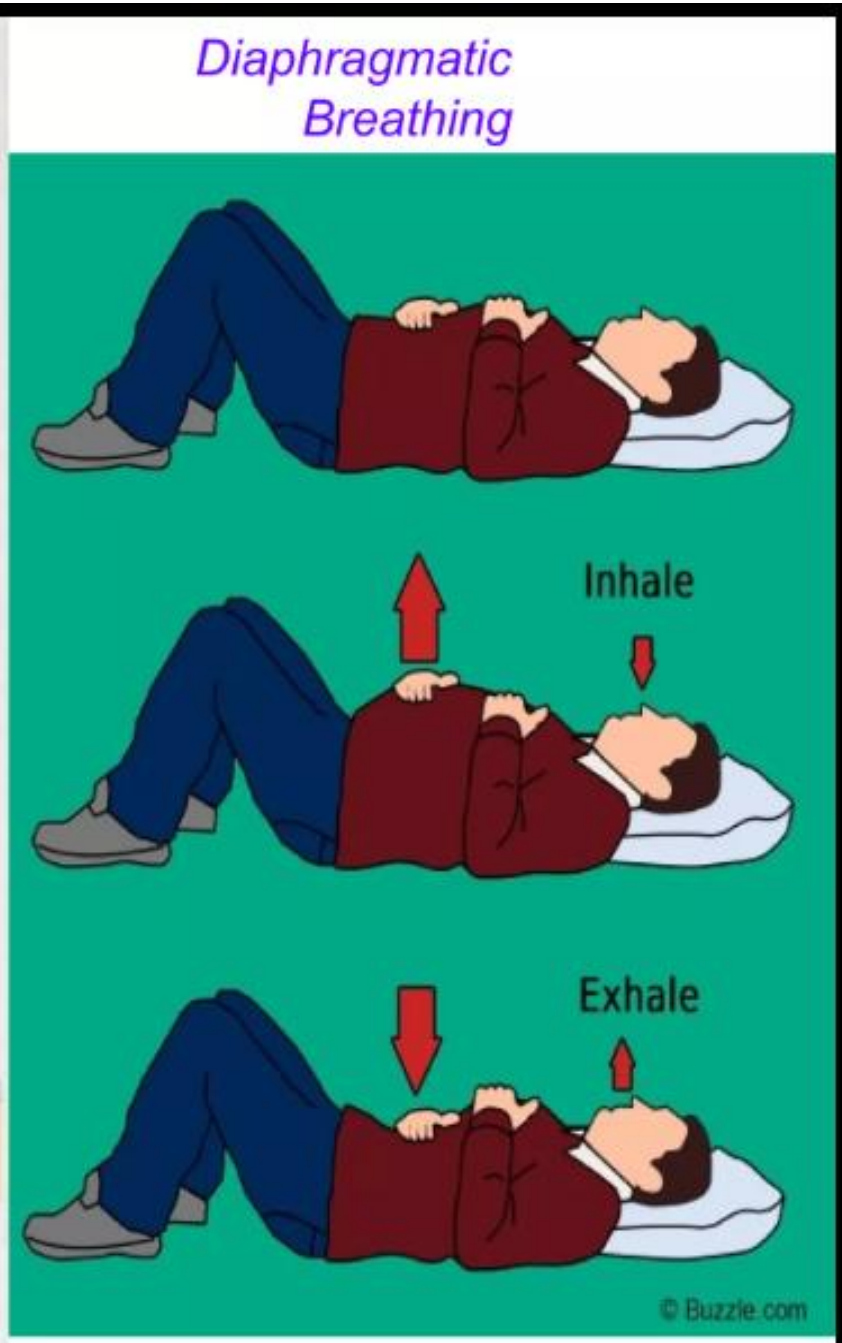
Bronchial Hygiene Techniques

- Postural drainage
- Percussion

→ Can be associated with others

- Directed cough: as Forced expiratory technique (huffing: small long (LL) or big short huff (UL) in cycle; 10 mins twice /d)
- Active cycle of breathing (breathing control (hands on abd.), deep breathing exercises e' breath hold (ribs) & huffing +/- manual technique)
- Autogenic drainage (self drainage: unstick, collect, evacuate)
- Positive expiratory pressure (behind mucus to push)
- Incentive Spirometry


Postural Drainage Positions
Upper Lobes
Apical
Segments
Right & left
Right & left
Anterior
segments
Right & left
Posterior
segments
Left
Right
Middle Lobes
Lingula
Right
Lower lobes
Basal
segments
Right & left:
Anterior
Right & left
posterior
Lateral
segments
Left
Right
Superior
segments
Right & left

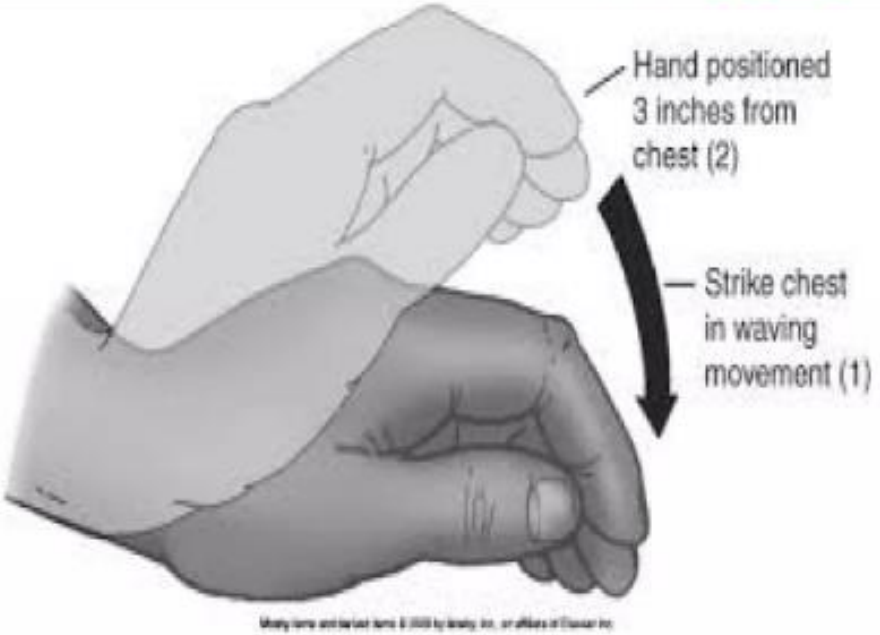

Diaphragmatic
Breathing
Inhale
Exhale
© Buzzle.com

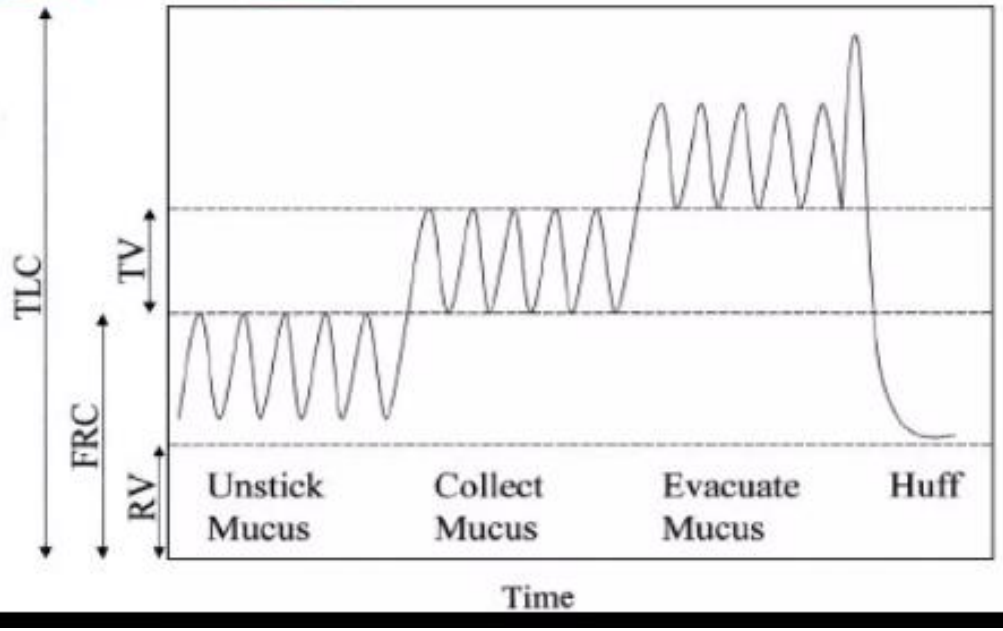
Vibratory PEP



Flutter device
Acapella





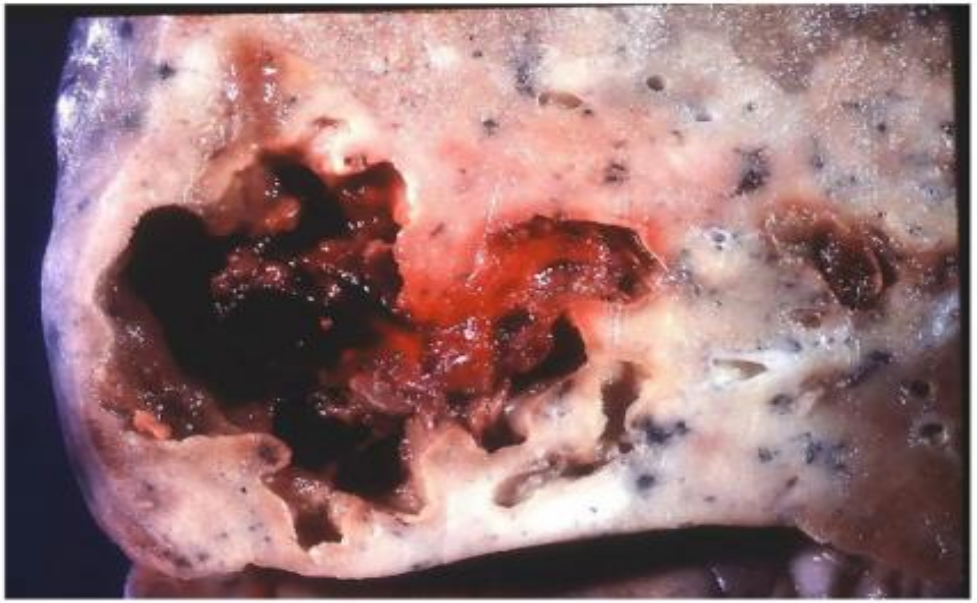


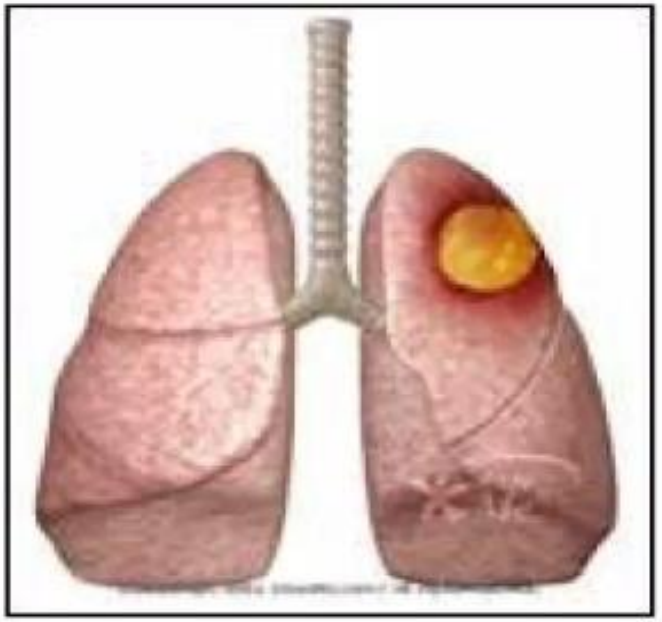


Lung Abscess

Definition:

- A lung abscess is a localized area of destruction of lung parenchyma in which infection by a Pyogenic organism results in tissue necrosis and suppuration and cavity formation.





Classification

Lung abscesses can be classified based on:

- **Duration:**
  - Acute abscesses are less than 4-6 wks old
  - Chronic abscesses are of longer duration
- **Number:** single or multiple
- **Etiology:**
  - Primary Abscess is infectious in origin: caused by aspiration or pneumonia in the healthy host.
  - Secondary Abscess
    - Pre-existing lung condition (obstruction, bronchiectasis, cyctic lung, c).
    - Hematogenous: Spread from an extra-pulmonary site (septic emboli)
    - Immuno-compromised state.
    - Inhalation of infected material
    - Necrotizing pneumonia (klebsiella, staph., pseudomonus, anearobic,

Pathogenesis:

- Lung abscess begin as an area of pneumonia → necrosis or microabscess → coalesce to form a single or sometimes multiple areas of suppuration → reach a size > 1cm in diameter → lung abscess.
- Inflammation erodes adjacent bronchi → suppuration is expectorated → air finds its way to the abscess cavity → fluid air interface.

Clinical features:

- **Presence of the predisposing factors**
- **Symptoms of acute lung abscess:**
  - The onset may be abrupt or gradual.
  - Fever with rigors, sweating, cough (dry cough at first which is followed by sudden onset of expectoration of large amounts of sputum after which fever drop).
  - Sputum: foul-odor, purulent with relation to posture.
  - Chest pain: it may be dull aching or severe pleuritic pain.
  - Hemoptysis may occur and may be massive.
  - Weight loss, anaemia and clubbing in chronic lung abscess (8-12 weeks).

Complications:

- Local spread to the same or to the other lung.
- Prolonged fever and chronicity.
- Massive hemoptysis.
- Residual bronchiectasis or fibrosis → Trapped lung
- Extension to the pleura or skin→ Empyema, BPF or PCF
- Metastatic brain abscess.
- Perinephric abscess.
- Amyloidosis.

Investigations:

1. Chest x-ray PA and lateral view

In the acute abscess the wall is thin with surrounding consolidation, and the chronic abscess the wall is thick.

2. Laboratory:

- **Sputum examination:**
  - Gram- stain film and culture (85% anaerobes)
  - Ziehl- Neelsen staining to exclude TB
  - Sputum cytology for malignant cells
  - Sputum culture for fungi and parasitic ova (if suspected)
- **Blood picture**: leucocytosis (>30.000/mm3) and anemia.

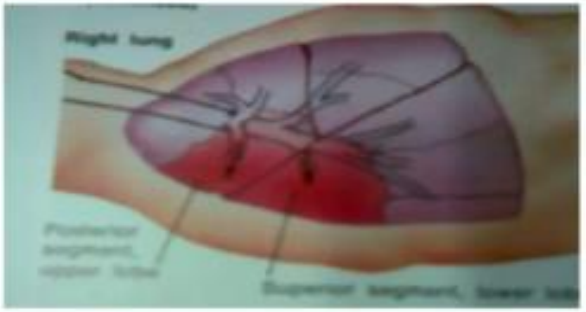
Irregularly shaped cavity with an air-fluid level inside

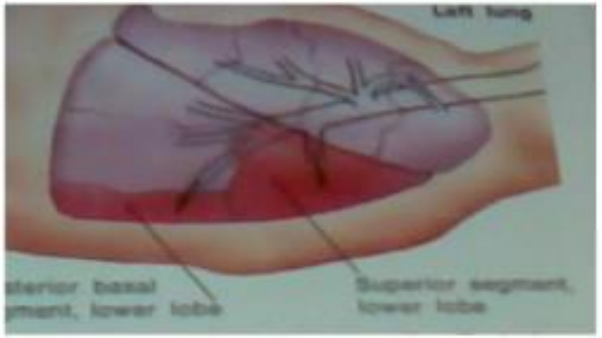
posterior segments of the upper lobes or

the superior segments of the lower lobes



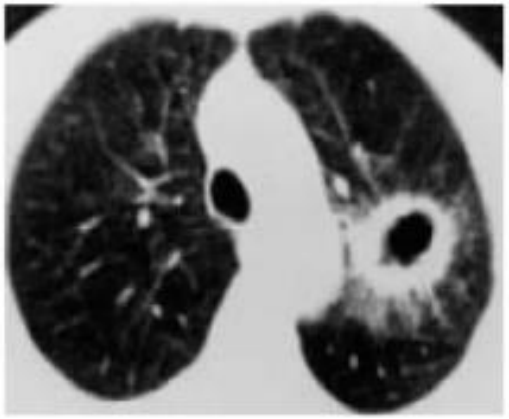












3. Bronchoscopy:

- **Diagnostic:** foreign body, tumors, or inspissated secretions.

- **Therapeutic:**

- Aspiration of secretions.
- Instillation of antibiotics.
- Foreign body extraction.
- Palliative treatment of bronchial carcinoma with laser or cryotherapy.

4. C-T chest: for suspected bronchial carcinoma.

Treatment:

1. *Prophylactic:* proper attention to the etiologic factors.

2. *Acute lung abscess:*

- Medical treatment is usually sufficient:

a) ***Antibiotics:*** 4 -6 wks → till resolution or small stable

- It is usually given empirically in the form of combined antibiotics to cover the spectrum of Gram positive, Gram negative and anaerobic organisms (Clindamycin).

b) ***Postural drainage*** is an important item in the management of lung abscess.

c) ***Bronchoscopic*** *aspiration* and instillation of antibiotics.

d) ***Symptomatic treatment:***

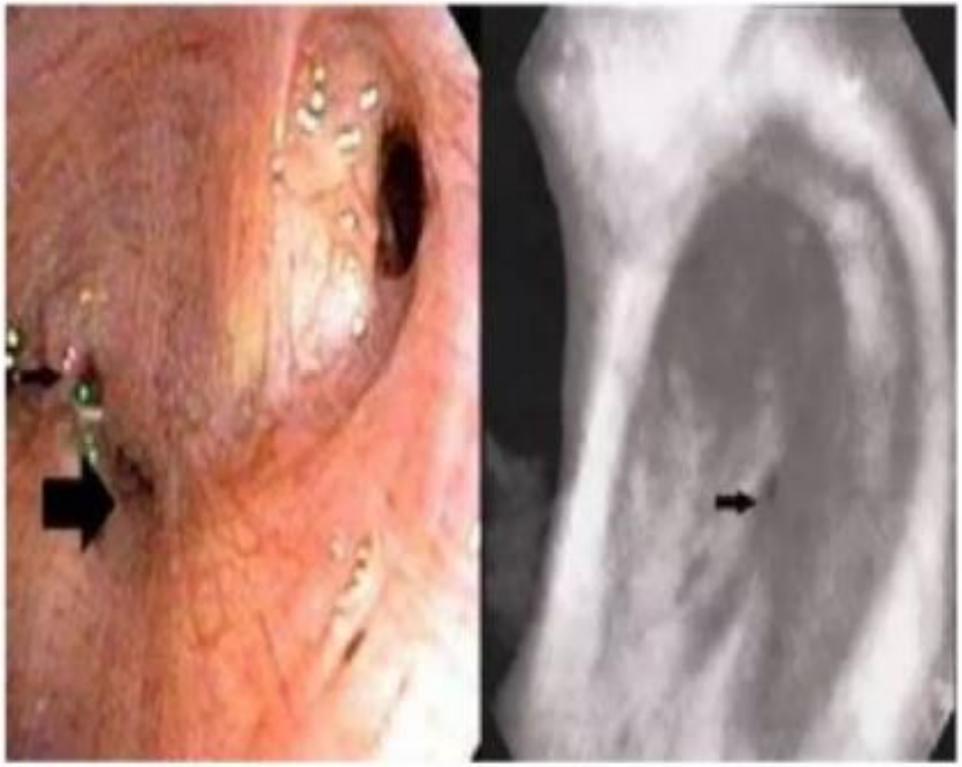
- Analgesics for pain.
- Expectorants.

e) ***Rest, good nutrition*** with ***high protein diet.***

3. Chronic lung abscess:

- Continue the medical treatment for another 6 wks.
- Surgical treatment (lobectomy or pneumonectomy) is indicated in chronic situation with:
  - Serious hemoptysis.
  - Failed medical treatment.
  - Suspected neoplasm
  - Congenital lung malformation

Empyema with Broncho-Pleural Fistula



Definition

- Pus in the pleural space or infected pleural fluid with fistula (opening) between the pleural space and the bronchial tree.

Causes:

- **Direct spread** from adjacent bacterial pneumonia.
- **Rupture** of a lung abscess into the pleural space.
- **Invasion** from subphrenic collection either pyogenic or amoebic.
- **Traumatic** penetration or **Iatrogenic**

  - It may be **acute** or **chronic (> 3 months)**.
  - It may be **loculated** or **free**.
  - It may be **post-operative** (2/3) or **non-operative** (1/3)

Clinical features

- Fever and chest pain.
- Dyspnea.
- Broncho-pleural fistula is characterized by postural cough and big amount of expectorated pus.
- Chronicity: pallor, malaise, weakness, easy fatigability, fever, anorexia and weight loss.
- Clubbing and pleural rub.
- Bubbling from chest tube

- **ACUTE**: sudden onset of dyspnoea,cough,expectoration of purulent material,hypotension,subcutn.emphysema,shifting of trachea and mediastinum.
- **SUBACUTE**: insidious onset of fever,malaise,wasting,minimally productive cough
- **CHRONIC**: associated with infectious disease,minimal mediastinal shift due to pleural and mediastinal fibrosis,not life threatening,adequate gas exchange in healthy lung
- Systemic features of sepsis

Investigations

- **Chest x-ray** (air-fluid level, tension pnx)
- **CT of the chest:** the condition of the underlying lung, Pnx
- **Thoracentesis:**
  - Foul - smelling aspirate (anaerobic infection).
  - Gram stain and culture and sensitivity: identification of the causative organisms.
  - Low PH (<7.2).
  - Pleural fluid white cell count > 15.000/ mm^3.
  - **Methylene blue test:** injection of methylene blue 1% in the pleural space, it will be expectorated in the sputum.
  - **Inhalation of radioactive isotopes:** detected in the pleural space.
  - **FOB:** Methylene blue test and visualization of bubbles after bronchial wash

Treatment:

- Appropriate antibiotic therapy.
- Intercostal tube drainage under water seal & pleurodesis
- Decortication
- Muscle flap closure of fistula
- Pleuropneumonectomy.

Case 1

⊙**Mr. K, male Patient, 25 years old, worker, no special habits.**

Complaint: Cough with expectoration of (usually) large quantities of offensive, purulent sputum, increased on lying on his back, lasting months in the past 7 years.

⊙Less frequently he has dyspnea, hemoptysis, fever, weakness and weight loss in the last 2 years

⊙There no relevant past or family history of similar condition, chronic sinusitis, GIT troubles.

Physical Examination

- General findings: digital clubbing, pallor, wasting, and weight loss.

- Chest examination crackles and rhonchi, heard on auscultation bilaterally.

- There is physical signs of cor pulmonale.



Clubbing

What is your provisional diagnosis?

- COPD
- Unresolved pneumonia
- Suppurative lung disease
- TB

What is your provisional diagnosis?

- COPD
- Unresolved pneumonia
- Suppurative lung disease
- TB

What is the most pathognomonic character in the history suggesting your diagnosis?

- The sputum odor
- Amount and color of the sputum
- Postural relation of the sputum
- Heamoptysis

What is the most pathognomonic character in the history suggesting your diagnosis?

- The sputum odor
- Amount and color of the sputum
- Postural relation of the sputum
- Heamoptysis

What is the significance of digital clubbing in this patient?

- Presence of cor pulmonale
- Presence of chronic hypoxemia
- Presence of chronic toxemia
- Persistent weight loss

What is the significance of digital clubbing in this patient?

- Presence of cor pulmonale
- Presence of chronic hypoxemia
- Presence of chronic toxemia
- Persistent weight loss

What do you suggest to support your diagnosis?

- Sputum culture and sensitivity
- Sputum analysis for acid fast bacilli
- Pulmonary function tests
- X ray chest and high resolution CT chest

What do you suggest to support your diagnosis?

- Sputum culture and sensitivity
- Sputum analysis for acid fast bacilli
- Pulmonary function tests
- X ray chest and high resolution CT chest

Laboratory Investigations

Tests include:

- Complete blood picture: anemia, leucocytosis with increase in PMNL.

- Sputum microbiological tests gram stain and culture is positive to Streptococci, Anaerobic bacteroids.

Chest X-Ray

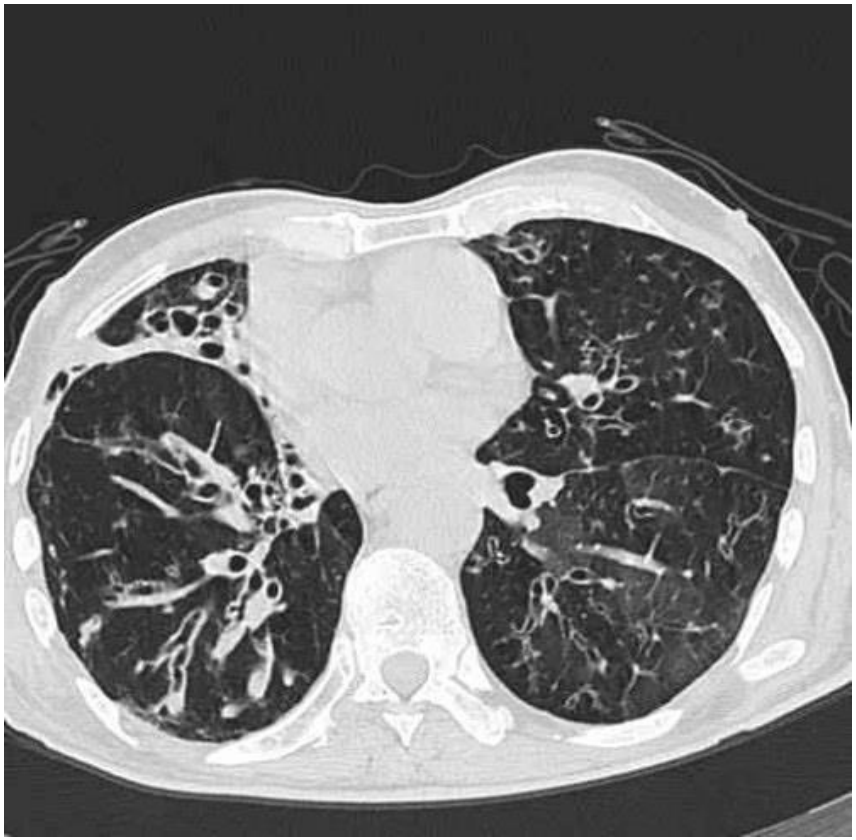
- It shows thickened bronchial walls, multiple cystic spaces

(Honeycombing)



Chest Computed Tomography (CT)

► HRCT is characterized by dilated bronchi with thickened walls bilaterally.





What is your final diagnosis?

- Post tuberculous bronchiectasis
- Kartagner syndrome
- Cystic fibrosis
- Acquired bronchiectasis secondry to childhood infections

What is your final diagnosis?

- Post tuberculous bronchiectasis
- Kartagner syndrome
- Cystic fibrosis
- Acquired bronchiectasis secondry to childhood infections

Culture in bronchiectasis usually reveal the following organisms except:

- Streptococcus pneumoniae
- Staphylococcus aureus
- Haemophilus influenza
- Pseudomonas aeruginosa
- Mycobacterium TB

Sputum Culture in bronchiectasis may reveal the following organisms except:

- Streptococcus pneumoniae
- Staphylococcus aureus
- Haemophilus influenza
- Pseudomonas aeruginosa
- Mycobacterium TB

NB: Sputum is usually offensive due to associated anaerobic infections e.g. Bacteroides

Pulmonary Function Tests in bronchiectasis may be?

- Non specific
- Limited diagnostic value
- Usually reveal an obstructive defect with concomitant restrictive defect
- All of the above

Question 3, Pulmonary Function Tests in bronchiectasis may be?

- Non specific
- Limited diagnostic value
- Usually reveal an obstructive defect with concomitant restrictive defect
- All of the above

Indications for resection for bronchiectasis include the following except:

1- Localized area (segment or lobe).
2- Severe or recurrent hemoptysis
3- Congenital bronchiectasis
4- Recurrent, severe episodes of infection

Indications for resection for bronchiectasis include the following except:

1- Localized area (segment or lobe).

2- Severe or recurrent hemoptysis

3- Congenital bronchiectasis

4- Recurrent, severe episodes of infection



Case 2

- A 16 years old male patients, student, single with no special habits.
- The patient has history of progressive dyspnea since childhood, cough and expectoration of large amount of yellow green sputum increased on lying on his back
- Past history of repeated chest infections, pneumothorax.
- No past history of TB, allergy, previous operations or GIT symptoms

General examination

- The patients looks ill, underweight, pale
- Finger nail clubbing
- Cardiac and abdominal examinations are free
- Mild lower limb edema

- Chest examination;
  - Bilateral medium sized to coarse crepitations all over the chest, scattered sonorous rhonchi.

Based on the history, and examination the patient may have the following except

- Congenital bronchiectasis
- Kartagner syndrome
- Infected cystic lung
- hypogammaglobulinemia

Based on the history, and examination the patient may have the following except

- Cystic fibrosis
- Kartagner syndrome
- Infected cystic lung
- hypogammaglobulinemia

How would you explain the lower limb edema in this patient?

- Cor pulmonale
- Nephrotic syndrome
- Liver cirrhosis
- Hypoproteinemia

How would you explain the lower limb edema in this patient?

- Cor pulmonale
- Nephrotic syndrome
- Liver cirrhosis
- Hypoproteinemia

X ray chest

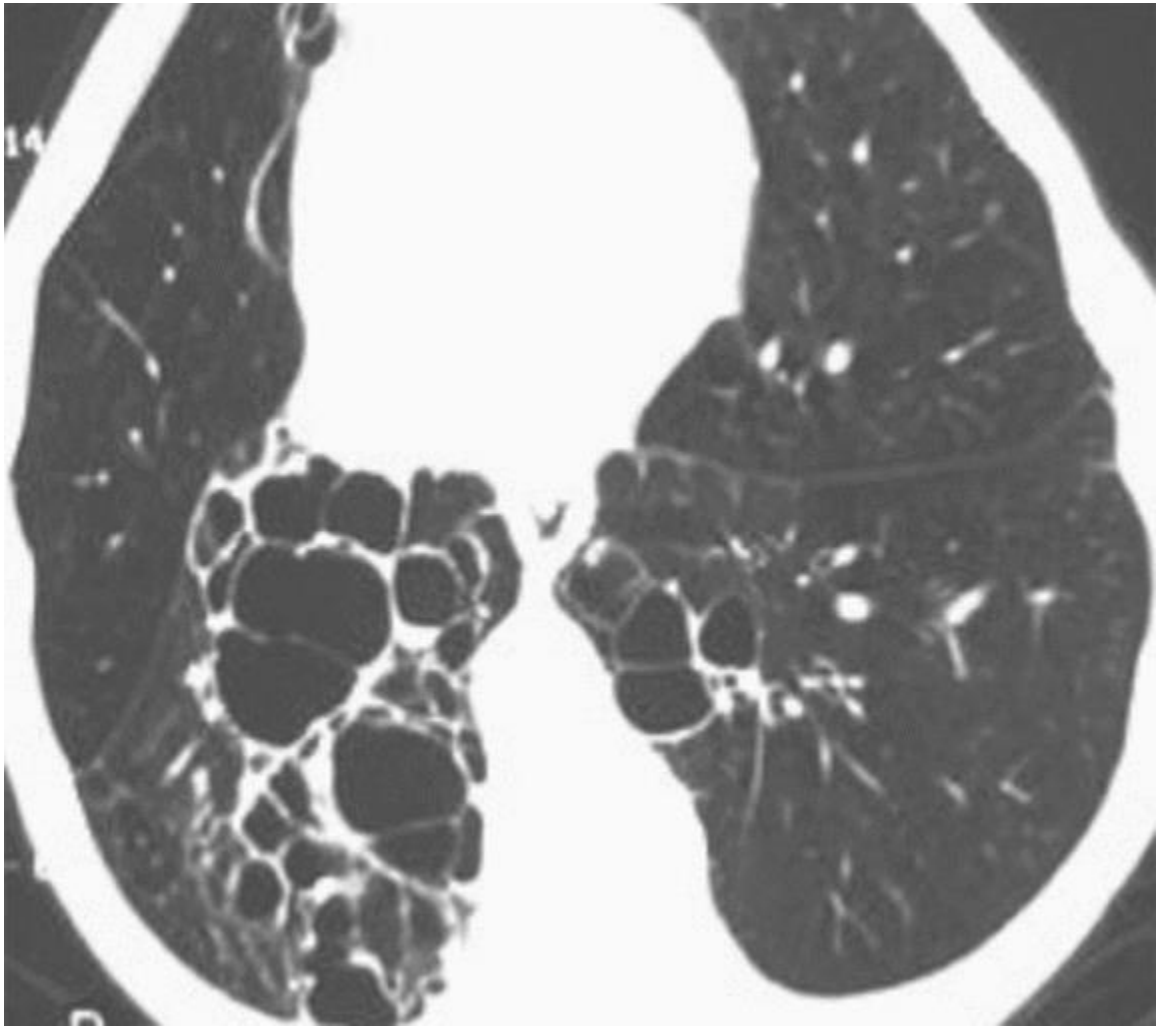
Diffuse bilateral air spaces (soap bubble appearance)





CT chest

bilateral thin walled cystic spaces in middle



Investigations

- Complete blood picture shows leucocytosis, mild anemia.
- ESR 23/39
- Echocardiography and abdominal sonography are free

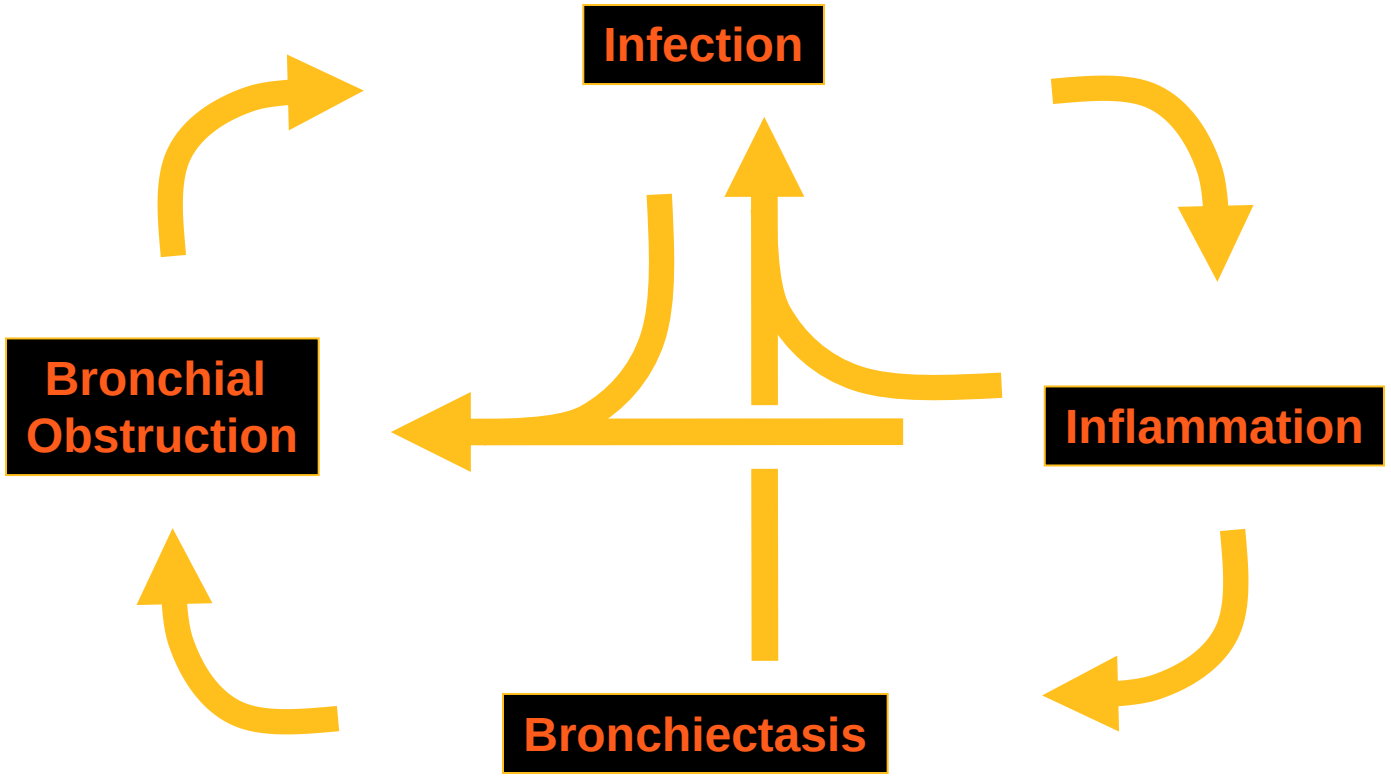
The main line of management of this case of infected cystic lung?

- Surgery
- Antibiotics
- Postural drainage
- Both postural drainage and antibiotic

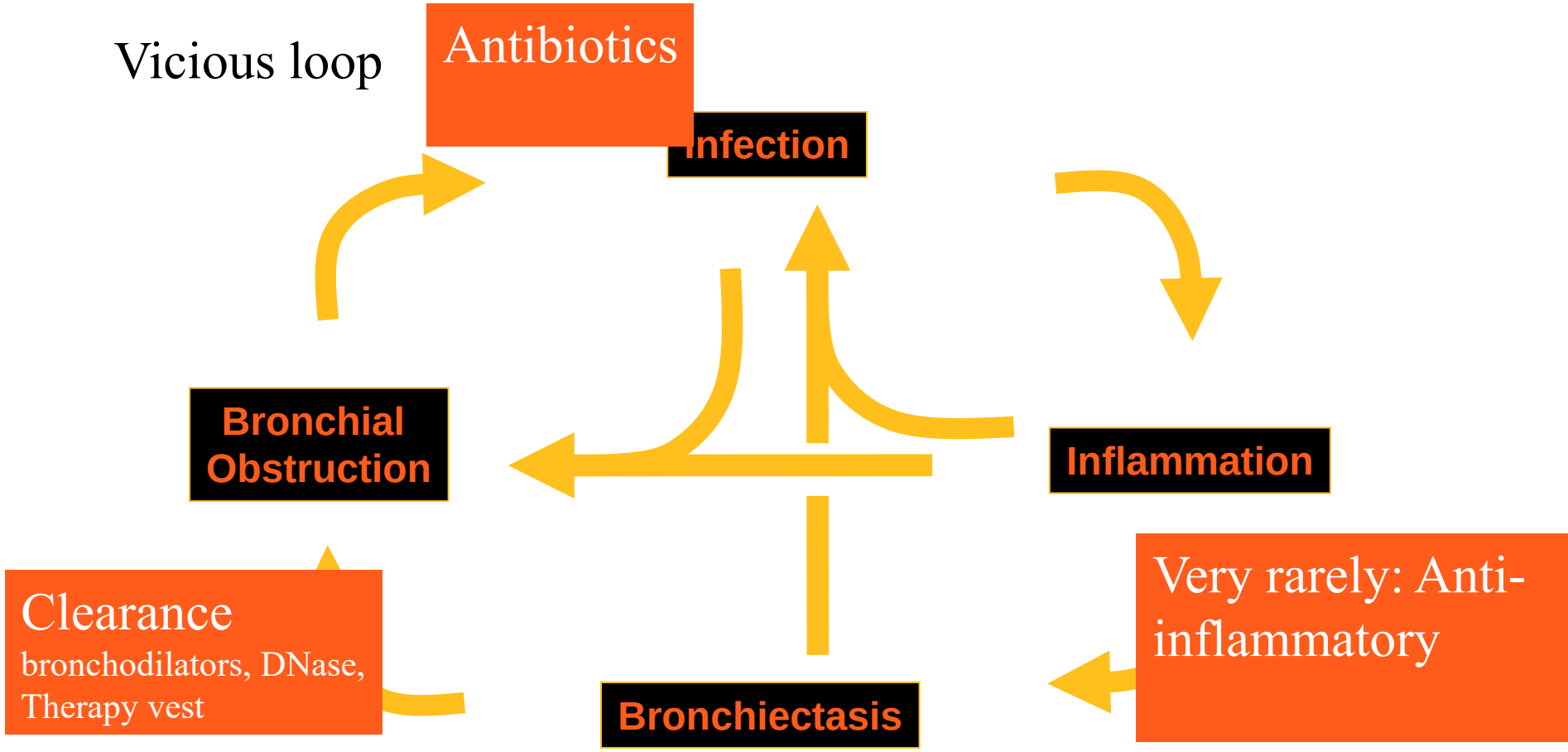
The main line of management of this case of infected cystic lung

- Surgery
- Antibiotics
- Postural drainage
- Both postural drainage and antibiotic

Infected cystic lung
(Congenital cystic adenomatoid malformation of the lung)



Vicious loop



Case 3

Complaint

► Female patient, 33 years old, house wife no special habits.

► **History** of fever, cough and expectoration of a big amount of sputum after upper abdominal operation (since 3 months).

► **Sputum** is usually purulent and foul odor.

Clinical examination

- Patient looks ill and feverish
- Clubbing
- Clinical signs of consolidation e.g. bronchial breathing and crepitations in the left infra-scapular area



Investigations

- Blood picture: increase in white cell count
- Sputum examination: mixed gram positive, negative and anaerobic organisms sensitive to different antibiotics.
- ESR 45/58

Plain X ray and CT chest





Chest X-ray: reveals rounded cavity with fluid level.
Computed tomography demonstrates the rounded cavity with regular thickened wall and air fluid level in the LLL

What is your provisional diagnosis?

- Sequestrated lobe
- Loculated empyema
- Lung abscess
- Unresolved pneumonia

What is your provisional diagnosis

- Sequestrated lobe
- Loculated empyema
- Lung abscess
- Unresolved pneumonia

Commonest cause of lung abscess is?

1- Aspiration
2- Post-traumatic lung abscess
3- Malignant lung abscess
4- Septic pulmonary embolization
5- Amoebic lung abscess

Commonest cause of lung abscess is?

1- Aspiration

2- Post-traumatic lung abscess

3- Malignant lung abscess

4**-** Septic pulmonary embolization

5- Amoebic lung abscess

These micro organisms may be recovered by culture from lung abscess except

- TB bacilli
- Respiratory syncetial virus
- Klebsiella pneumonea
- Anaerobic, peptostreptococcus bacteroids

These micro organisms may be recovered by culture from lung abscess except

- TB bacilli
- Respiratory syncetial virus
- Klebsiella pneumonea
- Anaerobic, peptostreptococcus bacteroids

Bronchoscopy in patients with suspected lung abscess is indicated in the following except:

1-When there is suspicion of foreign body inhalation

2- Endo-bronchial tumor

3- Tuberculosis

4- Installation of antibiotics

Bronchoscopy in patients with suspected lung abscess is indicated in the following except:

1- When there is suspicion of foreign body inhalation

2- Endo-bronchial tumor

3- Tuberculosis

4- Installation of antibiotics

Surgery is indicated in cases of lung abscess if:

- Failure of medical treatment
- Suspected foreign body
- Presence of bronchial adenoma
- All of the above

Surgery is indicated in cases of lung abscess if

- Failure of medical treatment
- Suspected foreign body
- Presence of bronchial adenoma
- All of the above

Case 4

Clinical Picture

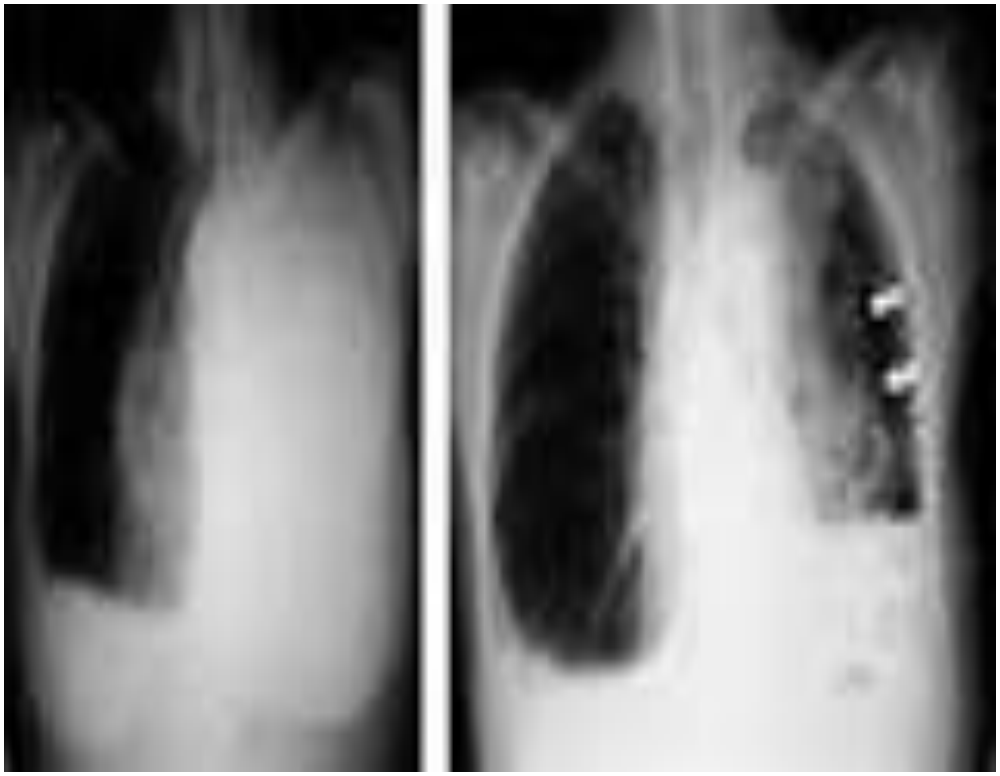
- Male patient 17 years old, student with no special habits.
- The patient has history of pneumonia with prolonged and recurrent fever, chest pain since 2 months.
- 2 weeks ago, the patient developed severe pleuretic chest pain on the left side, cough with big amount of expectorated pus increase on lying on the right.

Clinical examination

- Dyspnea
- High pyrexia with rigors, anemia, pallor, malaise
- Pleural pain and persistent cough
- Clinical signs of left sided pleural effusion (with shifting dullness)

1- Chest X-ray and CT Chest

- **pleural effusion (with fluid level)**.
- CT chest: left sided hydropneumothorax.





Investigations

2- Thoracentesis (Pleural taping)

►The pleural aspirate is purulent, foul-smelling.

►The pleural aspirate analyzed for the following:

- gram stain shows: mixed organisms, with positive aerobic and anaerobic cultures.
- pH is usually low (<7.2)
- Pleural fluid white cell count > 15.000/ml

3- Laboratory investigations

- PMN leucocytosis
- High ESR
- High C-reactive protein (CRP)

What is your provisional diagnosis?

- Bronchiectasis
- Lung abscess
- Empyema with broncho-pleural fistula
- Infected cystic lung

What is your provisional diagnosis?

- Bronchiectasis
- Lung abscess
- Empyema with broncho-pleural fistula
- Infected cystic lung

What is the diagnostic test for this case?

- X ray chest
- High resolution CT
- Methylene blue test
- Chest sonography

What is the diagnostic test for this case?

- X ray chest
- High resolution CT
- Methylene blue test
- Chest sonography

Indication of Ultrasound of the chest in this case is to define:

- Extent of pleural thickening
- Location of the fluid (general or loculated)
- To guide pleuracentesis (pleural tapping)
- All of the above

Indication of Ultrasound of the chest is to define:

- Extent of pleural thickening
- Location of the fluid (general or loculated)
- To guide pleuracentesis (pleural tapping)
- All of the above

The main line of treatment of empyema with bronchopleural fistula is?

- Drainage of empyema
- Use of antibiotic
- Fibrinolytic therapy
- Repair of the fistula

The main line of treatment of empyema with bronchopleural fistula is?

- Drainage of empyema
- Use of antibiotic
- Fibrinolytic therapy
- Repair of the fistula

Empyema with BP fistula

Definition

- It is pus in the pleural space or infected pleural fluid with fistula.

- The causative organism may or may not be isolated from the pus.

- Microscopically, neutrophil leucocytes are present in large number (>15. 000/ml).

Diagnosis of Bronchopleural Fistula

Injection of 1% of methylene blue in the pleural
cavity and to look for the blue dye in the sputum.

Pleural biopsy

May be needed to differentiate between tuberculous and non-tuberculous infections

Treatment

- Thoracentesis for easily aspirated pus by wide bored needle.
- Intercostal chest tube under water-seal drainage with initial antibiotic therapy till culture and sensitivity are available.
- Use of antibiotic for 2-4 weeks.
- Metronidazole with penicillin for anaerobic infection

Anti-tuberculous treatment for tuberculous empyema.

Empyema with BP fistula

Treatment

Surgery

- Cutting the pleural adhesions and freeing the loculi for optimal drainage
- Decortication for thickened pleura
- Surgical closure of bronchopleural fistula

Via Thoracotomy or Video-Assisted Thoracoscopic Surgery (VATS)


THANKYOU